



A multi-server retrial queueing-inventory system with asynchronous multiple vacations

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ABSTRACT

This study deals with asynchronous server vacation and customer retrial facilities in a multi-server queueing-inventory system. Customers arrive according to a Poisson process. The system comprises identical servers, a finite-size waiting area, and a storage area. The service time is distributed exponentially. If each server finds insufficient customers and items in the system after the busy period, they start a vacation. Once the server's vacation is over and it recognizes there is no chance of getting busy, it goes into an idle state; otherwise, it will take another vacation. Each server's vacation period occurs independently of the other servers. The system accepts a control policy for inventory replenishment. For the steady-state analysis, Neuts and Rao's matrix geometric approximation approach is used owing to the structure of an infinitesimal generator matrix. The necessary stability condition is computed. After calculating the sufficient system performance measures, an expected total cost of the system will be constructed and numerically incorporated with the parameters. Additional numerical analyses are conducted to examine customers' waiting time in the queue and orbit and the expected customer loss rate.

1. Introduction

The concept of a vacation in queueing system revolves around instances where a server becomes temporarily unavailable at a service station for a random duration. This period of absence, termed a "vacation", can arise from various circumstances, including deliberate decisions to allocate the server to a secondary service, such as maintenance or preparatory work for upcoming customers when the service station has no customers. This practice enables efficient utilization of the server's idle time for alternative tasks, thereby extending the applicability of queueing models to diverse real-world stochastic service setups. Since its introduction in Levy and Yechiali's paper [1], incorporating server vacations into queueing systems has garnered significant scholarly interest. For a comprehensive and complete review of vacation queueing systems, we recommend readers explore Doshi [2], Takagi [3], Shou Tian and Zhang [4], Ke et al. [5], and Chandrasekaran et al. [6].

Daniel and Ramanarayanan [7] introduced the vacation concept within a queueing-inventory system. They used "rest period" to refer to a server's vacation time. In [8], the server was allowed to take a break once the inventory count reached zero, and it was assumed that the

break duration followed a general distribution. Any customers who arrive during the server's break are regarded as lost. Narayanan et al. [9] studied service providing time in an (s, Q) queueing-inventory problem with vacation. Sivakumar [10] studied the multiple vacation feature with a retrial facility in the inventory system, where the server extends the vacation if no items are available in the inventory. Krishnamoorthy and Narayanan [11] studied the positive service time in the production inventory model with the vacation. Padmavathi et al. [12] considered two queueing-inventory models with two different types of vacations. In the first model, the server becomes idle when the inventory level hits zero. After this idle time, if the restocking has not occurred, the server enters a vacation known as a delayed vacation. When the server returns from vacation, the server becomes ready if the inventory is restocked. Otherwise, he takes one more vacation. In the second model, the server takes a single vacation when its inventory becomes stock-out.

Koroliuk et al. [13] examined the multiple vacations in a perishable queueing-inventory system and analyzed the system's performance asymptotically. Melikov et al. [14] examined the delayed vacation in an inventory system, assuming the server vacation while the inventory empties. If an inventory is restocked and the queue has pending tasks

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during this phase, the server becomes ready; otherwise, it enters a vacation after a random period. Vijaya Laxmi and Soujanya [15] studied a continuous review (s, S) inventory system operating with a service facility. After performing service on the item, the demanded item is issued to the customer. The server takes vacation based on the threshold level. Koroliuk et al. [16] investigated the queueing-inventory system server's vacation assignment. It can be noted that the server is allowed to take a break if no customers are waiting in the queue, and they also permit the server to terminate their vacation when the queue size reaches a predetermined threshold. Otherwise, his vacation time will be extended. Jeganathan et al. [17] obtained the minimal total cost for an (s, Q) inventory system with two heterogeneous servers, whereas only the second server can take a vacation. Suganya and Sivakumar [18] examined an inventory system with retrials involving two different servers and multiple vacations. The setup involves customers arriving based on a Markovian arrival process and the two parallel servers offering heterogeneous phase-type services to these customers. Yue and Qin [19] studied the production queueing-inventory system, assuming that the manufacturing facility would take a vacation when the current inventory level reaches its maximum capacity S . It terminates the vacation whenever the quantity of inventory items falls below s .

Jose and Beena [20] studied the vacation on considering the heterogeneous servers in a production inventory system with an orbit. In the vacation queueing-inventory model, Kathiresan and Anbazhagan [21] examined the two types of supply functions in which one has a minimal lead time than the other one. Jeganathan and Reiyas [22] elaborately studied the modified and delayed working vacations considering these two vacations to the two different servers. Beena and Jose [23] studied the multiple server vacation in two server production inventory systems. Both servers take a vacation whenever there is no inventory or customer in the system.

Manikandan and Nair [24] investigated the working vacation in an (s, Q) queueing-inventory model, which allows the server to work at a slower rate during the vacation time. Padmavathi and Sivakumar [25] created three models in which the server follows distinct vacation approaches: modified multiple vacation, single vacation, and multiple vacation policies within a queueing-inventory framework. Their investigation concluded that the modified multiple vacation policy outperforms others. Soujanya et al. [26] investigated the order placement with two suppliers instead of a single supplier in an inventory model with finite lifetimes of items and multiple working vacations. Krishnamoorthy et al. [27] conducted a study on a control strategy in which the server takes a break when a service procedure becomes inactive. This pause continues until sufficient participants are available to start the next service batch. While on break, the server generates items for upcoming services until it reaches a defined level L or the maximum service batch size, whichever comes first. Nithya et al. [28] explored the state-dependent arrival for a queueing-inventory model, assuming less arrival rate for the customers while the server is in a vacation state than the normal state. Radhamani et al. [29] showed that the variable-size ordering policy is more effective than the fixed and instantaneous ordering policy in a queueing-inventory model.

In a queueing-inventory system, Jeganathan et al. [30] studied the vacation for a flexible server for the two service stations. Zhang [31] contemplated multiple vacations for a queueing-inventory system server with the replenishment process under a random order size policy, and the lost sales during the server's absence were also observed. Zhang et al. [32] examined a queueing-inventory model with multiple vacation and impatient customers. They assumed that when the server is unavailable due to a vacation, newly arrived customers wait for service with a certain probability. Sugapriya et al. [33] analyzed an inventory system where the arrival of customers depends on the on-hand stock level, and the server takes a vacation when the stock level is zero. Jain and Kumar [34] used a Genetic algorithm to optimize the

cost function in a vacation queueing-inventory model considering two distinct suppliers for reordering items.

Recently, Jeganathan et al. [35] studied the server working in vacation mode combining the queue with the scrap items. Li et al. [36] considered an inventory model where the system allows the server to start vacation and reorder simultaneously once the inventory level drops to zero. Selvakumar et al. [37] studied the home delivery service process in the queueing-inventory model with vacation. He considered two servers: one remains in the service system, and another delivers items to the customer's doorstep. If the home delivery server finds no item to deliver while returning from the service, he takes a vacation. Ozkar [38] examined the production inventory model, in which when the inventory reaches the maximum level, the production facility takes a break for a random duration. Once the facility returns from the break, if the inventory drops to the predetermined level s , it has promptly turned on, and production resumes until the inventory level reaches S again. Yue et al. [39] studied vacation in a multi-server queueing-inventory system in which all the servers begin a vacation simultaneously whenever the on-hand inventory is empty. At the end of each vacation, if the on-hand inventory in the system is not empty, all servers observe that the inventory is no longer empty. All the servers return to the system to provide customer service at any time; otherwise, all the servers take one more vacation immediately and continue in the same way until the servers find the on-hand inventory is not empty. This vacation strategy is termed a "multiple synchronous vacation".

This extensive review of existing literature shows a research gap in the queueing-inventory systems that incorporate the concept of asynchronous multiple vacations. Few papers are available on the queueing models with asynchronous multiple vacations, which are given below.

Zhang and Tian [40] conducted a study on an asynchronous vacation queueing system in which the servers can take a vacation individually without customers in the queue. Moreover, the total number of servers on vacation must be at most a prefixed level of d during service or a vacation completion. Jain and Jain [41] considered a queueing system in which servers collectively engage in asynchronous vacations of random durations. This occurs when a specific number of servers become idle during service completion. However, the server can proceed with the vacation only if no customers are in the system. Wang et al. [42] analyzed customers' strategies to join an $M/M/K$ queue with both asynchronous and synchronous multiple vacation policies. They discovered that the equilibrium social welfare is greater under an asynchronous vacation approach when the traffic density is low. In contrast, the reverse is valid for a high traffic density under a synchronous vacation policy. Ma et al. [43] proposed a strategy that merges synchronous and asynchronous vacations. The scheduling approach is a two-dimensional Markov stochastic model involving multiple servers.

As stated earlier, through the extensive review of existing literature, we found a research gap in the queueing-inventory systems that incorporate the concept of asynchronous multiple vacations. To address this gap and enhance decision-making within the retail business, we integrate asynchronous multiple vacations within the queueing-inventory system.

The rest of the paper is arranged as follows: Section 2 deals with the model's notation and formulation. The solution of the assumed model is derived in Section 3, and its system performance measures are given in Section 4. In Section 5, the numerical analysis of the model is carried out, and finally, in the last section, the conclusion and the future direction are given.

2. Notations and model formulation

2.1. Notations

$0^*, Q^*$	: All servers are in vacation mode when the items are 0 and Q respectively
0	: Matrix containing all entries are zero
e	: Column vector of appropriate dimensions with each coordinate containing a value of one
I	: Identity matrix
δ_{ij}	: $\begin{cases} 1, & \text{if } j = i, \\ < 0, & \text{otherwise} \end{cases}$
$H(x)$	: $\begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{otherwise} \end{cases}$
λ	: Arrival rate of new customer
θ	: The rate at which a customer in orbit attempts to retry
μ	: The rate at which a customer receives service
η	: Vacation completion rate of a server
$\bar{\delta}_{ij}$	: $\begin{cases} 1 - \delta_{ij}, & \text{if } x = y = l, \\ 0, & \text{otherwise} \end{cases}$
$[D_l]_{x,y}$	: where D_l is a matrix of order l
$[E_l]_{x,y}$	: $\begin{cases} 0, & \text{if } x = y = l, \\ \theta, & \text{otherwise} \end{cases}$ where E_l is a matrix of order l
Ξ_a^b	: $\{a, a+1, \dots, b\}$ where a and b are integers
Ξ_a^∞	: $\{a, a+1, \dots\}$ where a is an integer
$P_1(t)$	: The number of customers in orbit at time t
$P_2(t)$	: The level of inventory at time t
$P_3(t)$	: The number of servers on vacation at time t
$P_4(t)$	: The number of servers busy with customers at time t
$P_5(t)$	: The number of idle servers at time t
$P_6(t)$	: The number of customers in the waiting hall at time t
v_1	: Holding cost per unit item per unit time
v_2	: Set up cost per order per unit time
v_3	: Waiting cost for a customer in the orbit per unit time
v_4	: Waiting cost for a customer in the queue per unit time
v_5	: Cost towards loss of a customer in the queue per unit time

Definition 2.1 (Asynchronous Vacation).[42] In the system, each server can take or terminate vacation individually; this is called asynchronous vacation.

2.2. Model formulation

The considered queueing-inventory system explores the performance of a multi-server service facility with asynchronous multiple vacations. Initially, the system has a maximum of S units of an item in the inventory storage space, there is a finite waiting hall of size N and a maximum of c identical servers who can provide service to the customer at an instant. When the system's current stock level falls to a prefixed reorder level of s , a reorder of $Q = (S - s)$ items is triggered at that epoch. This kind of reorder process is said to be an (s, Q) replenishment policy. The lead time follows an exponential distribution with the rate β . The relation between the number of servers and the reorder level is $c < s$. Upon completing a customer's service, one item will be deducted from the inventory storage space.

Depending on the system's circumstances, the server can be busy, idle, or on vacation.

(a) **Occurrence of Servers Busy Period:** Suppose the server is idle or completes the vacation or service. The busy period occurs in the following situations:

- While completing the service or vacation, if at least one item exists in the inventory (excluding servicing items) and at least one customer in the waiting hall (excluding servicing items), the server starts the service instantly.
- Suppose that the server is idle due to a lack of customers (excluding servicing customers) or items (excluding servicing items). The server becomes busy because of the customer's arrival or receiving the ordered items.

(b) **Occurrence of Servers Idle Period:** At the end of the service or vacation, if the server finds the following situations, the server goes to an idle state.

- There is at least one item (excluding the servicing items) in the inventory and no customer in the queue (excluding the servicing customers).
- There is at least one customer (excluding the servicing customers) in the queue and no item in the inventory (excluding the servicing items).

(c) **Occurrence of Servers Vacation Period:**

- The server goes into a vacation state at the end of each service if both customer and stock levels in the system (except servicing customers and servicing items) are zero.
- At the end of each vacation, if the server finds that both customer and stock levels in the system (except servicing customers and servicing items) are zero, then he takes one more vacation.

Customers arrive at the waiting hall according to a Poisson process with the rate λ . If the waiting room is at full capacity, a new customer either enters an infinite capacity orbit with probability p or leaves the system permanently with probability $1 - p$. This orbit is a virtual waiting area designed to minimize customer loss in the system. Under the classical retrial policy, any customer from the orbit can enter the waiting hall if the number of customers in the waiting hall is less than N . The retrial rate of a customer from the orbit is denoted as θ , and the inter-retrial times have an exponential distribution. A retrial can be successful or unsuccessful, depending on whether the number of customers in the queue is less than N or equal to N . The service time is distributed exponentially, and the service rate is defined by the parameter μ . The flow chart of the model is illustrated in Fig. 1.

3. Analysis and solutions of the model

Consider the six-dimensional Markov process

$$\{P(t) = (P_1(t), P_2(t), P_3(t), P_4(t), P_5(t), P_6(t)), t \geq 0\}$$

$$\text{with state space } \Omega = \bigcup_{i=1}^{11} B_i$$

$$B_1 = \{(i_1, 0, c - x_1, 0, x_1, i_6) \mid i_1 \in \Xi_0^\infty, x_1 \in \Xi_0^c, i_6 \in \Xi_{x_1}^N\}.$$

$$B_2 = \{(i_1, x_2 + 1, c - (j + 1), i_4, j + 1 - i_4, i_6) \mid i_1 \in \Xi_0^\infty, x_2 \in \Xi_1^{(c-2)}, j \in \Xi_0^{x_2-1}, i_4 \in \Xi_0^j\}.$$

$$B_3 = \{(i_1, x_2 + 1, c - (j + 1), j + 1, 0, i_6) \mid i_1 \in \Xi_0^\infty, x_2 \in \Xi_1^{(c-2)}, j \in \Xi_0^{x_2-1}, i_6 \in \Xi_{j+1}^N\}.$$

$$B_4 = \{(i_1, j, c - x_3, i_4, x_3 - i_4, i_6) \mid i_1 \in \Xi_0^\infty, x_3 \in \Xi_1^{(c-1)}, i_4 \in \Xi_0^{x_3-1}, j \in \Xi_c^S\}.$$

$$B_5 = \{(i_1, j, c - x_3, x_3, 0, i_6) \mid i_1 \in \Xi_0^\infty, x_3 \in \Xi_1^{(c-1)}, i_6 \in \Xi_{x_3}^N, j \in \Xi_c^S\}.$$

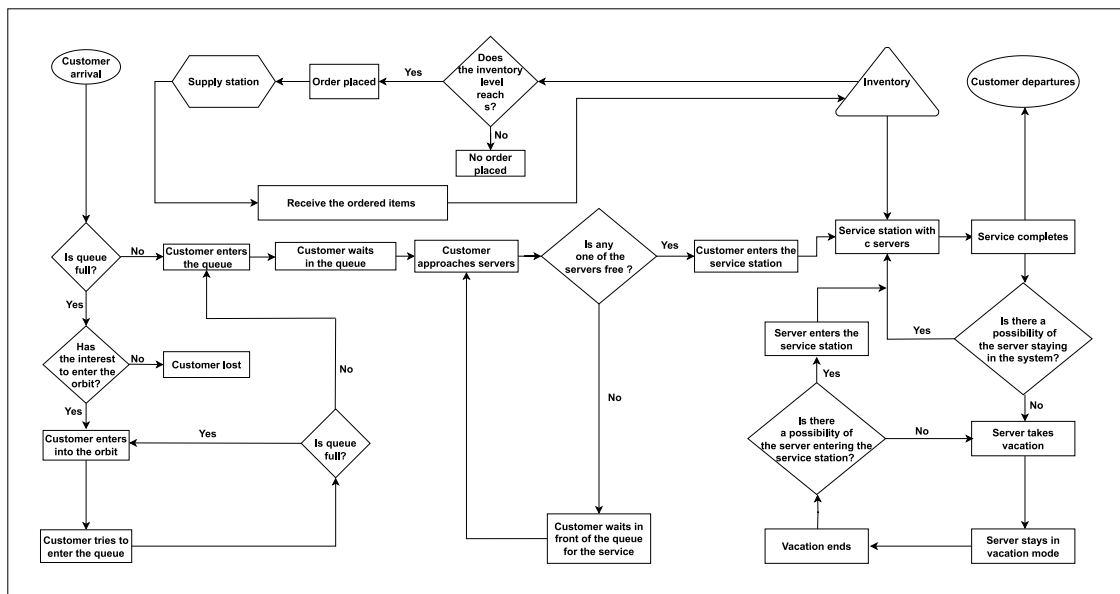


Fig. 1. Flow chart of the model.

$$B_6 = (l_1, Q, c, 0, 0, l_6) \mid l_1 \in \Xi_0^\infty, l_6 \in \Xi_0^N.$$

$$B_7 = (l_1, x_4 + 1, c - (x_4 + 1), l_4, x_4 + 1 - l_4, l_4) \delta$$

$$l_1 \in \Xi_0^\infty, x_4 \in \Xi_0^{(c-1)}, l_4 \in \Xi_0^{x_4}.$$

$$B_8 = \{ (l_1, x_5, c - x_5, x_5, 0, l_6) \mid l_1 \in \Xi_0^\infty, x_5 \in \Xi_1^{(c)}, l_6 \in \Xi_{x_5}^N \}.$$

$$B_9 = (l_1, x_6, c - (x_6 + j), x_6, j, l_6) \delta$$

$$l_1 \in \Xi_0^\infty, x_6 \in \Xi_1^{(c-1)}, j \in \Xi_1^{c-x_6}, l_6 \in \Xi_{(x_6+j)}^N.$$

$$B_{10} = (l_1, x_7, 0, l_4, c - l_4, l_4) \delta$$

$$l_1 \in \Xi_0^\infty, x_7 \in \Xi_{c+1}^{(S)}, l_4 \in \Xi_0^{(c-1)}.$$

$$B_{11} = (l_1, x_8, 0, c, 0, l_6) \mid l_1 \in \Xi_0^\infty, x_8 \in \Xi_{c+1}^{(S)}, l_4 \in \Xi_c^N.$$

3.1. Construction of infinitesimal generator matrix

Since $P_1(t)$ has an infinite number of levels (number of customers in orbit) and according to the classical retrial policy, the probability of a repeated attempt of an orbit customer is dependent on the orbit level. So that the process $\{P(t), t \geq 0\}$ has the following infinitesimal generator matrix is

$$H = \begin{pmatrix} H_{01} & H_0 & 0 & 0 & 0 & 5 \\ H_{10} & H_{11} & H_0 & 0 & 0 & 5 \\ 0 & H_{20} & H_{21} & H_0 & 0 & 5 \\ 4 & 4 & 4 & 4 & 4 & 7 \end{pmatrix} \quad (1)$$

From the structure of the matrix H , it is clear that the process $\{P(t), t \geq 0\}$ is the level-dependent quasi-birth-death process. The stationary distribution of this process is also dependent on the orbit level. It can be explicitly stated in [Theorem 4](#). All the matrices in the above infinitesimal generator matrix H have dimension

$$Sc(N+1) + (c+2)N - (2c^3 + 3c^2 - 5c - 12)(1/6).$$

1. The non-zero upper diagonal matrix, H_0 describes the transitions of the new arrival to the orbit if the waiting hall is full.

$$\mathbf{H}_0 = \text{diag}(\mathbb{k}_{0*}, \mathbb{k}_{0*}, \mathbb{k}_0, \mathbb{k}_1, \dots, \mathbb{k}_{c-1}, \mathbb{k}_c, \mathbb{k}_c, \dots, \mathbb{k}_c), \quad (2)$$

where

$$\mathbb{k}_j = \text{diag}(D_{N-(c-1)}, \dots, D_{N-j}, D_{N+1}, \dots, D_{N+1}), \quad (3)$$

$$j \in \Xi_{\circ}^{(c)}$$

is a sub-block matrix of order c .

2. The non-zero lower diagonal matrix, H_{i10} denotes the transitions of orbit customers enter the waiting hall.

For $l_1 \in \mathcal{E}_1^\infty$,

$$\mathbf{H}_{i,0} = \text{diag}(\mathbf{W}_{0^*}, \mathbf{W}_{Q^*}, \mathbf{W}_0, \dots, \mathbf{W}_{c-1}, \mathbf{W}_c, \dots, \mathbf{W}_c), \quad (4)$$

where

$$\mathbf{W}_j = \text{diag}(E_{N-(c-1)}, \dots, E_{N-j}, E_{N+1}, \dots, E_{N+1}), \quad (5)$$

$$j \in \Xi_0^{(c)},$$

is a sub-block matrix of order c .

3. The diagonal matrix, H_{i1} describes the transitions of primary arrival, service process, reorder process, server taking a vacation and vacation completion process.

For $l_1 = 0, 1, 2, \dots$

$$H_{i_1 i_2} = \begin{array}{lll} \text{h} & J'_{i_2} & i'_2 = 0^*, \quad i_2 = 1, \\ \text{n} & J'_{i_2} & i'_2 = Q^*, 0, \quad i_2 = 0^*, \\ \text{n} & J'_{i_2} & i'_2 = Q, \quad i_2 = Q^*, \\ \text{n} & F_{i_2} & i'_2 = i_2 - 1, \quad i_2 \in \Xi_1^c, \\ \text{n} & F_{c+1} & i'_2 = i_2 - 1, \quad i_2 \in \Xi_{(c+1)}^S, \\ \text{n} & L_{i_2} & i'_2 = i_2 + Q, \quad i_2 \in \Xi_c^c, \\ \text{n} & L_c & i'_2 = i_2 + Q, \quad i_2 \in \Xi_{(c+1)}^S, \\ \text{n} & G'_{i_2} & i'_2 = i_2, \quad i_2 \in 0^*, Q^*, \\ \text{n} & G'_{i_2} & i'_2 = i_2, \quad i_2 \in \Xi_0^{(-1)}, \\ \text{n} & G_c & i'_2 = i_2, \quad i_2 \in \Xi_c^S, \\ \text{n} & 0 & \text{otherwise.} \end{array}$$

The sub-matrices of $\mathbf{H}_{i+1}$ are given in the [Appendix A](#).

The ergodicity of the Markov chain $P(t)$ is assumed to $\rho = \frac{\lambda}{c\mu} < 1$ (See [44]).

3.2. Neuts and Rao's truncation method

Neuts and Rao's truncation method [45] is developed for the analysis of the level-dependent quasi-birth-death process by appropriate truncation level in orbit. So, the level-dependent quasi-birth-death process is converted into a level-independent quasi-birth-death process from the truncation point. That is, by setting $H_{i,0} = H_{M,0}$ and $H_{i,1} =$

H_{M1} for all $t_1 \in M$, the equilibrium of the system is determined. So the modified generator matrix of $\{P(t), t \in 0\}$ is given by

$$\hat{H} = \begin{array}{cccccccccccc} \begin{array}{c} \backslash \\ r \\ r \\ r \\ r \\ r \\ r \\ p \end{array} & \begin{array}{c} H_{01} \\ H_{10} \\ 0 \\ 4 \\ 0 \\ 0 \\ 0 \\ 4 \end{array} & \begin{array}{c} H_0 \\ H_{11} \\ H_{20} \\ 4 \\ 0 \\ 0 \\ 0 \\ 4 \end{array} & \begin{array}{c} 0 \\ H_0 \\ H_{21} \\ 4 \\ 0 \\ 0 \\ 5 \\ 4 \end{array} & \begin{array}{c} 5 \\ 0 \\ 0 \\ 4 \\ 5 \\ 0 \\ 0 \\ 4 \end{array} & \begin{array}{c} 0 \\ 0 \\ 0 \\ 4 \\ H_{M0} \\ H_{M0} \\ 4 \\ 4 \end{array} & \begin{array}{c} 0 \\ 0 \\ 0 \\ 4 \\ H_{M1} \\ H_{M1} \\ 4 \\ 4 \end{array} & \begin{array}{c} 0 \\ 0 \\ 0 \\ 4 \\ H_0 \\ H_0 \\ 4 \\ 4 \end{array} & \begin{array}{c} 5 \\ 5 \\ 5 \\ 4 \\ 0 \\ 5 \\ 5 \\ 4 \end{array} & \begin{array}{c} a \\ s \\ s \\ s \\ s \\ s \\ s \\ q \end{array} \end{array}$$

Theorem 1. The probability vector ϕ in steady-state, which is related to the finite generator matrix H_M at truncation point M , can be obtained from the following equation: $H_M \phi = 0$, where $H_M = H_{M0} + H_{M1} + H_0$ and the vector ϕ as converted as follows:

$$\phi^{(i_2)} = \phi^{(Q)} \nabla_{i_2}, \quad i_2 = 0^*, Q^*, 0, 1, \dots, S. \quad (6)$$

where

$$\nabla_{l_2} = \begin{cases} \begin{matrix} h & (-1)^Q (F_{c+1} \hat{G}_c^{-1})^{Q-c} \\ n & \pm_{i=1}^{c-1} F_{i+1} \hat{G}_i^{-1} J_{l_2} \hat{G}_{l_2}^{-1}, \\ n & (-1)^{Q+1} (F_{c+1} \hat{G}_c^{-1})^{Q-c} \\ n & \pm_{i=1}^{c-1} F_{i+1} \hat{G}_i^{-1} J_{0^*} \hat{G}_{0^*}^{-1} J_{l_2} \hat{G}_{l_2}^{-1}, \end{matrix} & l_2 = 0^*, \\ \begin{matrix} h & (-1)^Q (F_{c+1} \hat{G}_c^{-1})^{Q-c} \\ n & \pm_{i=1}^{c-1} F_{i+1} \hat{G}_i^{-1} J_{l_2} \hat{G}_{l_2}^{-1}, \\ n & (-1)^{Q+1} (F_{c+1} \hat{G}_c^{-1})^{Q-c} \\ n & \pm_{i=1}^{c-1} F_{i+1} \hat{G}_i^{-1} J_{0^*} \hat{G}_{0^*}^{-1} J_{l_2} \hat{G}_{l_2}^{-1}, \end{matrix} & l_2 = Q^*, \end{cases}$$

[illegible]

$$\nabla_{I_2} = \begin{vmatrix} (-1)^{Q-(j-1)}(F_{c+1}\hat{G}_c^{-1})^{Q-s} \\ {}^3_c \\ x=j+1 & (F_{c+1}\hat{G}_c^{-1})^{s-c} & \pm k & F_i\hat{G}_{i-1}^{-1} \\ L_3\hat{G}_c^{-1}(F_{c+1}\hat{G}_c^{-1})^{x-(j+1)} + \\ x=0 & (F_{c+1}\hat{G}_c^{-1})^{s-c-x}L_c\hat{G}_c^{-1}(F_{c+1}\hat{G}_c^{-1})^x & I_2 \in Q+j, \\ (F_{c+1}\hat{G}_c^{-1})^{c-j}, & j \in \Xi_1^{(c-1)}, \end{vmatrix}$$

$$\nabla_{i_2} = \begin{cases} \frac{(-1)^{Q-(j-1)}(F_{c+1}\hat{G}_c^{-1})^{Q-s}}{\sum_{x=0}^{s-j}(F_{c+1}\hat{G}_c^{-1})^{s-j-x}L_c\hat{G}_c^{-1}(F_{c+1}\hat{G}_c^{-1})^x} \% \\ 0, & \begin{array}{l} i_2 \in Q + j, \\ j \in \Xi_{c+1}, \\ \text{otherwise,} \end{array} \end{cases}$$

and $\phi^{(Q)}$ is obtained by solving the equations

$$\phi^{(Q^*)} J_{l_2} + \phi^{(0)} L_0 + \phi^{(l_2)} \hat{G}_c + \phi^{(l_2+1)} F_{c+1} = 0 \quad (7)$$

and

$$\sum_{l_2=0^*, Q^*}^E \phi^{(l_2)} e + \sum_{l_2=0}^E \phi^{(l_2)} e = 1. \quad (8)$$

Proof. We have

$$\phi H_M = 0 \text{ and } \phi e = 1, \quad (9)$$

where

$$H_M{}_{i_2, i_2'} = \begin{cases} J_{i_2'}', & i_2' = 0^*, & i_2 = 1, \\ J_{i_2'}', & i_2' = Q^*, 0, & i_2 = 0^*, \\ J_{i_2'}', & i_2' = Q, & i_2 = Q^*, \\ F_{i_2}, & i_2' = i_2 - 1, & i_2 \in \Xi_c^{-1}, \\ F_{c+1}, & i_2' = i_2 - 1, & i_2 \in \Xi_{c+1}^S, \\ L_{i_2}, & i_2' = i_2 + Q, & i_2 \in \Xi_c^0, \\ L_c, & i_2' = i_2 + Q, & i_2 \in \Xi_{(c+1)}^S, \\ \hat{G}_{i_2}, & i_2' = i_2, & i_2 \in 0^*, Q^*, \\ \hat{G}_{i_2}, & i_2' = i_2, & i_2 \in \Xi_0^{(c-1)}, \\ \hat{G}_c, & i_2' = i_2, & i_2 \in \Xi_c^S, \\ 0, & \text{otherwise.} \end{cases}$$

The sub-matrices $\hat{\mathbf{G}}_{l'_2}$ for $l'_2 = 0^*, Q^*$, and $l'_2 \in \Xi_0^c$ are given in the [Appendix B](#).

The equation $\phi H_M = 0$ yields the following set of equations:

$$\phi^{(0*)}\hat{G}_{l_2} + \phi^{(1)}J_{l_2} = 0, \text{ if } l_2 = 0^*, \quad (10)$$

$$\phi^{(0^*)} \mathbf{J}_{l_2} + \phi^{(Q^*)} \hat{\mathbf{G}}_{l_2} = 0, \text{ if } l_2 = Q^* \quad (11)$$

$$\phi^{(0*)}J_{l_2} + \phi^{(0)}\hat{G}_{l_2} + \phi^{(1)}F_{l_2+1} = 0, \text{ if } l_2 = 0 \quad (12)$$

$$\phi^{(l_2)} \hat{G}_{l_2} + \phi^{(l_2+1)} F_{l_2+1} = 0, \text{ if } l_2 \in \Xi_1^c \quad (13)$$

$$\phi^{(l_2)} \hat{G}_c + \phi^{(l_2+1)} F_{c+1} = 0, \text{ if } l_2 \in \Xi_{c+1}^{(Q-1)} \quad (14)$$

$$\phi^{(Q^*)} \mathbf{J}_b + \phi^{(0)} \mathbf{L}_0 + \phi^{(i_2)} \hat{\mathbf{G}}_c + \phi^{(i_2+1)} \mathbf{F}_{c+1} = 0, \text{ if } i_2 = Q \quad (15)$$

$$\phi^{(i_2-Q)}L_{i_2-Q} + \phi^{(i_2)}\hat{G}_c + \phi^{(i_2+1)}F_{c+1} = 0, \text{ if } i_2 \in \Xi_{Q+1}^{(Q+c-2)} \quad (16)$$

$$\phi^{(l_2-Q)}L_c + \phi^{(l_2)}\hat{G}_c + \phi^{(l_2+1)}F_{c+1} = 0, \text{ if } l_2 \in \Xi_{0+c-1}^{(S-1)} \quad (17)$$

$$\phi^{(l_2-Q)} \mathbf{L}_c + \phi^{(l_2)} \hat{\mathbf{G}}_c = 0, \text{ if } l_2 = S \quad (18)$$

We arrive at the stated solution by iteratively solving the aforementioned system of equations with its normalization condition $\phi_e = 1$.

It should be noted that the associated state space of the modified infinitesimal generator matrix $\hat{H}$ is the same as H . However the ergodicity condition of the original Markov chain $\lambda < c\mu$ does not hold for the truncated Markov chain. Because the retrial rate is constant from the level M up. So following the general theory for the Quasi-Birth-and-Death Process, the ergodicity condition of the truncated Markov chain is derived as given in [Theorem 2](#).

Theorem 2. *The stability condition for the modified infinitesimal generator matrix, $\hat{H}$ is given by the inequality*

$$z_1 p \lambda < z_2 M \theta, \quad (19)$$

where,

$$z_1 = \phi^{(0,c,0,0,N)} + \phi^{(Q,c,0,0,N)} + \sum_{i_2=0}^{c-1} \sum_{i_3=0}^{c-1-i_2} \phi^{(i_2,i_3,i_2,c-(i_2+i_3),N)} \\ + \sum_{i_2=1}^{c-1} \sum_{i_3=c-i_2}^{c-1} \phi^{(i_2,i_3,c-i_3,0,N)} + \sum_{i_2=c}^{c-1} \sum_{i_3=c}^{c-1} \sum_{i_4=0}^{c-1-i_2-i_3} \phi^{(i_2,i_3,i_4,c-(i_3+i_4),N)}$$

and

$$z_2 = \sum_{i_6=0}^{N-1} \phi^{(0,c,0,0,i_6)} + \sum_{i_6=0}^{N-1} \phi^{(Q,c,0,0,i_6)} \\ + \sum_{i_7=0}^{c-1} \sum_{i_3=0}^{-(c+1)} \sum_{i_6=c-i_3}^{N-1} \phi^{(i_2,i_3,i_2,c-(i_2+i_3),i_6)}$$

$$\begin{aligned}
& \binom{c-1}{i_2=0} \binom{c-1}{i_3=c-i_2} \binom{c-1}{i_4=0} \phi^{(i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \binom{c-1}{i_2=0} \binom{c-1}{i_3=c-i_2} \binom{N-1}{i_6=c-i_3+1} \phi^{(i_2, i_3, i_2, 0, i_6)} \\
& + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{c-1}{i_4=0} \phi^{(i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{N-1}{i_6=c-i_3+1} \phi^{(i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

Proof. According to Marcel F Neuts [46], the steady-state probability vector Φ for the modified infinitesimal generator matrix exists if and only if the following inequality is satisfied:

$$\phi H_0 e < \phi H_{M0} e.$$

Now, writing all the matrices and a stationary vector to the generator matrix H_M is given by

$$\begin{aligned}
& p\lambda\phi^{(0,c,0,0,N)} + p\lambda\phi^{(Q,c,0,0,N)} \\
& + \binom{c-1}{i_2=0} \binom{c-1}{i_3=0} p\lambda\phi^{(i_2, i_3, i_2, c-(i_2+i_3), N)} \\
& + \binom{c-1}{i_2=1} \binom{c-1}{i_3=c-i_2} p\lambda\phi^{(i_2, i_3, c-i_3, 0, N)} \\
& + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{c-1}{i_4=0} p\lambda\phi^{(i_2, i_3, i_4, c-(i_3+i_4), N)} \\
& < \binom{N-1}{i_6=0} M\theta\phi^{(0,c,0,0,i_6)} + \binom{N-1}{i_6=0} M\theta\phi^{(M,Q,c,0,0,i_6)} \\
& + \binom{c-1}{i_2=0} \binom{c-1}{i_3=0} \binom{N-1}{i_6=c-i_3} M\theta\phi^{(i_2, i_3, i_2, c-(i_2+i_3), i_6)} \\
& + \binom{c-1}{i_2=0} \binom{c-1}{i_3=c-i_2} \binom{N-1}{i_6=c-i_3+1} M\theta\phi^{(i_2, i_3, i_4, c-(i_3+i_4), i_4)} \\
& + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{N-1}{i_6=c-i_3+1} M\theta\phi^{(i_2, i_3, i_2, 0, i_6)} \\
& + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{N-1}{i_6=c-i_3+1} M\theta\phi^{(i_2, i_3, i_4, c-(i_3+i_4), i_4)} \\
& + \binom{c-1}{i_2=c} \binom{c-1}{i_3=0} \binom{N-1}{i_6=c-i_3+1} M\theta\phi^{(i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

Hence, we get the desired inequality (19).

3.3. Limiting probability distribution

The generator matrix H has a steady state probability vector $\Phi = (\Phi^{(0)}, \Phi^{(1)}, \Phi^{(2)}, \dots)$, and it satisfies the stability condition. Thus, the Markov process

$$(P_1(t), P_2(t), P_3(t), P_4(t), P_5(t), P_6(t)), t \geq 0$$

with state space Ω is regular. Consequently, the limiting probability distribution is given by:

$$\Phi^{(i_1, i_2, i_3, i_4, i_5, i_6)} =$$

$$\lim_{t \rightarrow \infty} Pr P_1(t) = i_1, P_2(t) = i_2, P_3(t) = i_3,$$

$$P_4(t) = i_4, P_5(t) = i_5, P_6(t) = i_6$$

$$P_1(0) = 0, P_2(0) = 0, P_3(0) = 0,$$

$$P_4(0) = 0, P_5(0) = 0, P_6(0) = 0.$$

exists and is independent of the initial state.

3.4. Computation of R-matrix

Theorem 3. Let the matrix R can be determined by the matrix quadratic equation

$$R^2 H_{M0} + R H_{M1} + H_0 = 0. \quad (20)$$

The square matrix R is of dimension

$$Sc(N+1) + (c+2)N - \frac{2c^3 + 3c^2 - 5c - 12}{6}$$

and is defined by

$$\begin{aligned}
& \begin{matrix} & 0^* & Q^* & 0 & 5 & S \\ 0^* & \diagdown A^{(0^*, 0^*)} & A^{(0^*, Q^*)} & A^{(0^*, 0)} & 5 & A^{(0^*, S)} \\ Q^* & \diagup A^{(Q^*, 0^*)} & A^{(Q^*, Q^*)} & A^{(Q^*, 0)} & 5 & A^{(Q^*, S)} \\ 0 & \diagup A^{(0, 0^*)} & A^{(0, Q^*)} & A^{(0, 0)} & 5 & A^{(0, S)} \\ R=1 & \diagup A^{(1, 0^*)} & A^{(1, Q^*)} & A^{(1, 0)} & 5 & A^{(1, S)} \\ 4 & \diagup 4 & 4 & 4 & 5 & 4 \\ S-1 & \diagup A^{(S-1, 0^*)} & A^{(S-1, Q^*)} & A^{(S-1, 0)} & 5 & A^{(S-1, S)} \\ S & \diagup A^{(S, 0^*)} & A^{(S, Q^*)} & A^{(S, 0)} & 5 & A^{(S, S)} \end{matrix} \quad (21)
\end{aligned}$$

If $i_2 = 0^*, Q^*$ and $i'_2 = 0^*, Q^*$

$$\begin{aligned}
& \begin{matrix} \diagdown & 0 & 0 & 5 & 0 & a \\ \diagup & 0 & 0 & 5 & 0 & s \\ R & \diagup & 4 & 4 & 4 & s \\ & \diagup & e_0^{i_2, i'_2} & e_1^{i_2, i'_2} & 5 & e_N^{i_2, i'_2} \end{matrix} \quad q_{(N+1) \times (N+1)}
\end{aligned}$$

If $i_2 = 0^*, Q^*$ and $i'_2 = 0$

$$A^{(i_2, i'_2)} = x_0^{i_2, i'_2}, x_1^{i_2, i'_2}, x_2^{i_2, i'_2}, 5, x_{c-1}^{i_2, i'_2}.$$

If $i_3 \in \Xi_0^{(c-1)}$

$$\begin{aligned}
& \begin{matrix} \diagdown & 0 & 0 & 5 & 0 & a \\ \diagup & 0 & 0 & 5 & 0 & s \\ x_{i_3}^{i_2, i'_2} & \diagup & 4 & 4 & 4 & s \\ & \diagup & e_1^{i_2, i'_2} & e_2^{i_2, i'_2} & 5 & e_{N-(c-(i_3+1))}^{i_2, i'_2} \end{matrix} \quad q_{(N+1) \times (N-(c-(i_3+1)))}
\end{aligned}$$

If $i_2 = 0^*, Q^*$ and $i'_2 \in \Xi_1^{(c-1)}$

$$A^{(i_2, i'_2)} = x_0^{i_2, i'_2}, x_1^{i_2, i'_2}, x_2^{i_2, i'_2}, 5, x_{c-1}^{i_2, i'_2}.$$

If $i_3 \in \Xi_0^{(c-(i_2+1))}$

$$\begin{aligned}
& \begin{matrix} \diagdown & 0 & 0 & 5 & 0 & a \\ \diagup & 0 & 0 & 5 & 0 & s \\ x_{i_3}^{i_2, i'_2} & \diagup & 4 & 4 & 4 & s \\ & \diagup & e_1^{i_2, i'_2} & e_2^{i_2, i'_2} & 5 & e_{N-(c-(i_3+1))}^{i_2, i'_2} \end{matrix} \quad q_{(N+1) \times (N-(c-(i_3+1)))}
\end{aligned}$$

If $i_3 \in \Xi_{(c-(i_2+1)+1)}^{(c-1)}$

$$\begin{aligned}
& \begin{matrix} \diagdown & 0 & 0 & 5 & 0 & a \\ \diagup & 0 & 0 & 5 & 0 & s \\ x_{i_3}^{(i_2, i'_2)} & \diagup & 4 & 4 & 4 & s \\ & \diagup & e_0^{i_2, i'_2} & e_1^{i_2, i'_2} & 5 & e_N^{i_2, i'_2} \end{matrix} \quad q_{(N+1) \times (N+1)}
\end{aligned}$$

If $i_2 = 0^*, Q^*$ and $i'_2 \in \Xi_c^S$

$$A^{(i_2, i'_2)} = x_0^{i_2, i'_2}, x_1^{i_2, i'_2}, x_2^{i_2, i'_2}, 5, x_{c-1}^{i_2, i'_2}.$$

If $i_3 \in \Xi_0^{c-1}$

$$\begin{aligned}
& \begin{matrix} \diagdown & 0 & 0 & 0 & 0 & a \\ \diagup & 0 & 0 & 0 & 0 & s \\ x_{i_3}^{i_2, i'_2} & \diagup & 4 & 4 & 4 & s \\ & \diagup & e_0^{i_2, i'_2} & e_1^{i_2, i'_2} & 5 & e_N^{i_2, i'_2} \end{matrix} \quad q_{(N+1) \times (N+1)}
\end{aligned}$$

If $i_2 = 0$ and $i'_2 = 0^*, Q^*$

$$\begin{aligned}
& \begin{matrix} b & x_0^{i_2, i'_2} & c \\ f & x_1^{i_2, i'_2} & g \\ f & x_1^{i_2, i'_2} & g \\ f & x_2^{i_2, i'_2} & g \\ f & 5 & g \\ f & x_{c-1}^{i_2, i'_2} & g \\ d & x_{c-1}^{i_2, i'_2} & e \end{matrix}
\end{aligned}$$

If $t_3 \in \Xi_0^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 0 & 0 & 0 & 0 & s \\ r & 4 & 4 & 4 & 4 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N-(c-(t_3+1))) \times (N+1)} \end{matrix}$$

If $t_2 \in \Xi_1^{(c-1)}$ and $t'_2 = 0^*, Q^*$

$$A^{(t_2, t'_2)} = \begin{matrix} b & x_0^{t_2, t'_2} & c \\ f & x_0^{t_2, t'_2} & g \\ f & x_1^{t_2, t'_2} & g \\ f & x_1^{t_2, t'_2} & g \\ f & x_2^{t_2, t'_2} & g \\ f & 5 & g \\ f & x_2^{t_2, t'_2} & g \\ f & x_{c-1}^{t_2, t'_2} & g \\ d & x_{c-1}^{t_2, t'_2} & e \end{matrix}$$

If $t_3 \in \Xi_0^{(c-(t_2+1))}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 5 & 0 & a \\ r & 0 & 0 & 5 & 0 & s \\ r & 4 & 4 & 4 & 4 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N-(c-(t_3+1))) \times (N+1)} \end{matrix}$$

If $t_3 \in \Xi_{(c-(t_2+1)+1)}^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 0 & 0 & 0 & 0 & s \\ r & 4 & 4 & 4 & 4 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N+1) \times (N+1)} \end{matrix}$$

If $t_2 \in \Xi_c^S$ and $t'_2 = 0^*, Q^*$

$$A^{(t_2, t'_2)} = \begin{matrix} b & x_0^{t_2, t'_2} & c \\ f & x_0^{t_2, t'_2} & g \\ f & x_1^{t_2, t'_2} & g \\ f & x_1^{t_2, t'_2} & g \\ f & x_2^{t_2, t'_2} & g \\ f & 5 & g \\ f & x_2^{t_2, t'_2} & g \\ f & x_{c-1}^{t_2, t'_2} & g \\ d & x_{c-1}^{t_2, t'_2} & e \end{matrix}$$

If $t_3 \in \Xi_0^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 0 & 0 & 0 & 0 & s \\ r & 4 & 4 & 4 & 4 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N+1) \times (N+1)} \end{matrix}$$

If $t_2 = 0$ and $t'_2 = 0$

$$A^{(t_2, t'_2)} = \begin{matrix} \backslash & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & a \\ r & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & s \\ r & x_{1,0}^{t_2, t'_2} & x_{1,1}^{t_2, t'_2} & 5 & x_{1,c-1}^{t_2, t'_2} & s \\ r & 4 & 4 & 4 & 4 & s \\ r & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & s \\ p & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & q_{c \times c} \end{matrix}$$

If $t_3 \in \Xi_0^{(c-1)}$ and $t'_3 \in \Xi_0^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_1 & e_2 & \dots & e_N & q_{(N-(c-(t_3+1))) \times (N-(c-(t'_3+1)))} \end{matrix}$$

If $t_2 \in \Xi_1^{(c-1)}$ and $t'_2 \in \Xi_1^{(c-1)}$

$$A^{(t_2, t'_2)} = \begin{matrix} \backslash & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & a \\ r & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & s \\ r & x_{1,0}^{t_2, t'_2} & x_{1,1}^{t_2, t'_2} & 5 & x_{1,c-1}^{t_2, t'_2} & s \\ r & 4 & 4 & 4 & 4 & s \\ r & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & s \\ p & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & q_{c \times c} \end{matrix}$$

If $t_3 \in \Xi_0^{(c-(t_2+1))}$ and $t'_3 \in \Xi_0^{(c-(t'_2+1))}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_1 & e_2 & \dots & e_N & q_{(N-(c-(t_3+1))) \times (N-(c-(t'_3+1)))} \end{matrix}$$

If $t_3 \in \Xi_0^{(c-(t_2+1))}$ and $t'_3 \in \Xi_{(c-(t'_2+1)+1)}^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N-(c-(t_3+1))) \times (N+1)} \end{matrix}$$

If $t_3 \in \Xi_{(c-(t_2+1)+1)}^{(c-1)}$ and $t'_3 \in \Xi_0^{(c-(t'_2+1))}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_1 & e_2 & \dots & e_N & q_{(N+1) \times (N-(c-(t'_3+1)))} \end{matrix}$$

If $t_3 \in \Xi_{(c-(t_2+1)+1)}^{(c-1)}$ and $t'_3 \in \Xi_{(c-(t'_2+1)+1)}^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & e_2 & \dots & e_N & q_{(N+1) \times (N+1)} \end{matrix}$$

If $t_2 \in \Xi_c^S$ and $t'_2 \in \Xi_c^S$

$$A^{(t_2, t'_2)} = \begin{matrix} \backslash & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & a \\ r & x_{0,0}^{t_2, t'_2} & x_{0,1}^{t_2, t'_2} & 5 & x_{0,c-1}^{t_2, t'_2} & s \\ r & x_{1,0}^{t_2, t'_2} & x_{1,1}^{t_2, t'_2} & 5 & x_{1,c-1}^{t_2, t'_2} & s \\ r & 4 & 4 & 4 & 4 & s \\ r & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & s \\ p & x_{c-1,0}^{t_2, t'_2} & x_{c-1,1}^{t_2, t'_2} & 5 & x_{c-1,c-1}^{t_2, t'_2} & q_{c \times c} \end{matrix}$$

If $t_3 \in \Xi_0^{(c-1)}$ and $t'_3 \in \Xi_0^{(c-1)}$

$$x_{t_3}^{t_2, t'_2} = \begin{matrix} \backslash & 0 & 0 & 0 & 0 & a \\ r & 4 & 4 & 4 & 4 & s \\ r & 0 & 0 & 0 & 0 & s \\ r & t_2, t'_2 & t_2, t'_2 & \dots & e_N^{t_2, t'_2} & s \\ p & e_0 & e_1 & \dots & e_N & q_{(N+1) \times (N+1)} \end{matrix}$$

Proof. By utilizing the matrices H_{M0} , H_{M1} , and H_0 in Eq. (20), we can derive a set of nonlinear equations as follows:

If $t_2 = 0^*, Q^*$, $t_2 \in \Xi_0^S$ and $t'_2 = 0^*$

$$\begin{aligned} \dot{E} \sum_{j=0^*, Q^*} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \dot{E} \sum_{j=0} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \\ A^{(t_2, t'_2)} G_{t'_2} + A^{(t_2, 1)} J_{t'_2} + \delta_{t_2 t'_2} B_{t'_2} = 0. \end{aligned} \quad (22)$$

If $t_2 = 0^*, Q^*$, $t_2 \in \Xi_0^S$ and $t'_2 = Q^*$

$$\begin{aligned} \dot{E} \sum_{j=0^*, Q^*} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \dot{E} \sum_{j=0} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \\ A^{(t_2, t'_2)} G_{t'_2} + A^{(t_2, 0^*)} J_{t'_2} + \delta_{t_2 t'_2} B_{t'_2} = 0. \end{aligned} \quad (23)$$

If $t_2 = 0^*, Q^*$, $t_2 \in \Xi_0^S$ and $t'_2 = 0$

$$\begin{aligned} \dot{E} \sum_{j=0^*, Q^*} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \dot{E} \sum_{j=0} A^{(t_2, j)} A^{(j, t'_2)} W_{t'_2} + \\ A^{(t_2, t'_2)} G_{t'_2} + A^{(t_2, 0^*)} J_{t'_2} + A^{(t_2, t'_2+1)} F_{t'_2+1} + \delta_{t_2 t'_2} B_{t'_2} = 0. \end{aligned} \quad (24)$$

If $i_2 = 0^*, Q^*, i_2 \in \Xi_0^S$ and $i_2' \in \Xi_1^c$

$$\sum_{j=0^*, Q^*} \dot{E} A^{(i_2, j)} A^{(j, i_2')} W_{i_2'} + \sum_{j=0} \ddot{E} A^{(i_2, j)} A^{(j, i_2')} W_{i_2'} + A^{(i_2, i_2')} G_{i_2'} + A^{(i_2, i_2'+1)} F_{i_2'+1} + \delta_{i_2 i_2'} B_{i_2'} = 0. \quad (25)$$

If $i_2 = 0^*, Q^*, i_2 \in \Xi_0^S$ and $i_2' \in \Xi_{c+1}^{Q-1}$

$$\sum_{j=0^*, Q^*} \dot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + \sum_{j=0} \ddot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + A^{(i_2, i_2')} G_c + A^{(i_2, i_2'+1)} F_{c+1} + \delta_{i_2 i_2'} B_c = 0. \quad (26)$$

If $i_2 i_2' = 0^*, Q^*, i_2 \in \Xi_0^S$ and $i_2' = Q$

$$\sum_{j=0^*, Q^*} \dot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + \sum_{j=0} \ddot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + A^{(i_2, i_2')} G_c + A^{(i_2, Q^*)} J_{i_2'} + A^{(i_2, Q-i_2')} L_{Q-i_2'} + A^{(i_2, i_2'+1)} F_{c+1} + \delta_{i_2 i_2'} B_c = 0. \quad (27)$$

If $i_2 i_2' = 0^*, Q^*, i_2 \in \Xi_0^S$ and $i_2' \in \Xi_{Q+1}^{Q+c}$

$$\sum_{j=0^*, Q^*} \dot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + \sum_{j=0} \ddot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + A^{(i_2, i_2')} G_c + A^{(i_2, Q-i_2')} L_{Q-i_2'} + A^{(i_2, i_2'+1)} F_{c+1} + \delta_{i_2 i_2'} B_c = 0. \quad (28)$$

If $i_2 i_2' = 0^*, Q^*, i_2 \in \Xi_0^S$ and $i_2' \in \Xi_{Q+c+1}^S$

$$\sum_{j=0^*, Q^*} \dot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + \sum_{j=0} \ddot{E} A^{(i_2, j)} A^{(j, i_2')} W_c + A^{(i_2, i_2')} G_c + A^{(i_2, Q-i_2')} L_c + A^{(i_2, i_2'+1)} F_{c+1} + \delta_{i_2 i_2'} B_c = 0. \quad (29)$$

solving Eqs. (22) to (29) by Gauss-Seidel iterative method, we will obtain the R matrix.

Theorem 4. The stationary probability vector $\Phi^{(i_1)}$ of the Markov chain can be determined by

$$\Phi^{(i_1)} = \begin{cases} \Phi^{(0)} \Omega_{i_1} & \text{if } i_1 \in \Xi_1^M \\ \Phi^{(0)} \Omega_M R^{(i_1-M)} & \text{if } i_1 > M \end{cases} \quad (30)$$

Where R is obtained from Eq. (20), it can be obtained from Theorem 3,

$$\Omega_{i_1} = \begin{cases} H_0 K_j & \text{if } i_1 \in \Xi_1^M \\ \sum_{j=0}^{i-1} [- (H_0 K_{j+1} H_{j+2,0} + H_{j+1,1})]^{-1} & \text{if } j \in \Xi_1^{(M-1)} \\ [- (H_{j,1} + R H_{j,0})]^{-1} & \text{if } j = M \end{cases}$$

and $\Phi^{(0)}$ is obtained by solving the system of equations

$$\Phi^{(0)} (H_{0,1} + \Omega_1 H_{1,0}) = 0. \quad (31)$$

$$\sum_{i_1=0} \dot{E} \Phi^{(i_1)} e = 1. \quad (32)$$

Proof. let $\Phi = (\Phi^{(0)}, \Phi^{(1)}, \Phi^{(2)}, \dots)$ be a probability vector which satisfies

$$\Phi \hat{H} = 0 \text{ and } \Phi e = 1. \quad (33)$$

Solving the $\Phi \hat{H} = 0$, we get the following system of equations

$$\Phi^{(0)} H_{0,1} + \Phi^{(1)} H_{1,0} = 0. \quad (34)$$

$$\Phi^{(i-1)} H_0 + \Phi^{(i)} H_{i,1} + \Phi^{(i+1)} H_{i+1,0} = 0, \quad \forall i \in \Xi_1^{(M-1)} \quad (35)$$

$$\Phi^{(M-1)} H_0 + \Phi^{(M)} (H_{M,1} + R H_{M,0}) = 0. \quad (36)$$

$$\text{and } \sum_{i_1=0}^{M-1} \dot{E} \Phi^{(i_1)} + \Phi^{(M)} (1 - R)^{-1} e = 1. \quad (37)$$

Solving Eqs. (35) to (36), we obtain Eq. (30) for the values $i_1 \in \Xi_1^M$. Assume that

$$\Phi^{(i_1)} = \Phi^{(M)} R^{(i_1-M)} \text{ if } i_1 \in \Xi_M^\infty, \quad (38)$$

substitute $\Phi^{(M)}$ values in Eq. (38) again we obtain Eq. (30) for $i_1 > M$. From (34), we get (31). Substitute Eq. (30) in (32) and solving Eqs. (31) and (32) simultaneously we get $\Phi^{(0)}$.

4. System performance metrics

Using the steady-state probability vector, various system performance metrics are defined in this section as follows:

1. The expected inventory level (i_1) of the multi-server queueing-inventory system in the steady-state is defined by

$$i_1 = \sum_{i_1=0} \dot{E} \sum_{i_6=0} \ddot{E} Q \Phi^{(i_1, Q, c, 0, 0, i_6)} + \sum_{i_1=0} \dot{E} \sum_{i_2=1} \ddot{E} i_2 \Phi^{(i_1, i_2)} e.$$

2. Under the (s, Q) reordering policy, an order is immediately placed when the inventory level drops to s at the end of the service. Using the vector Φ , we can figure out the expected reorder rate (i_2) of the multi-server queueing-inventory system:

$$i_2 = \sum_{i_1=0} \dot{E} \sum_{i_3=0} \ddot{E} \sum_{i_4=1} \ddot{E} i_4 \mu \Phi^{(i_1, s+1, i_3, i_4, c-(i_3+i_4), i_4)} + \sum_{i_1=0} \dot{E} \sum_{i_3=0} \ddot{E} (c-i_3) \mu \Phi^{(i_1, s+1, i_3, c-i_3, 0, i_6)}.$$

3. The expected number of customers in the orbit is denoted by i_3 . When arriving customers discover the waiting hall is full, they enter the orbit with probability p . At any given time, the expected number of orbital customers present is

$$i_3 = \sum_{i_1=0}^{M-1} \dot{E} i_1 \Phi^{(i_1)} e + M (I - R)^{-1} \Omega_M \Phi^{(0)} e.$$

4. When arriving customers find that the waiting hall has reached its maximum level, they enter the orbit under the Bernoulli schedule with probability p . The expected rate at which the customer enters the orbit is denoted by i_4 and is given by

$$i_4 = \sum_{i_1=0} \dot{E} p \lambda \Phi^{(i_1, 0, c, 0, 0, N)} + \sum_{i_1=0} \dot{E} p \lambda \Phi^{(i_1, Q, c, 0, 0, N)} + \sum_{i_1=0} \dot{E} \sum_{i_2=0} \ddot{E} \sum_{i_3=0} \ddot{E} p \lambda \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), N)} + \sum_{i_1=0} \dot{E} \sum_{i_2=1} \ddot{E} \sum_{i_3=c-i_2} \ddot{E} p \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, N)} + \sum_{i_1=0} \dot{E} \sum_{i_2=c} \ddot{E} \sum_{i_3=0} \ddot{E} p \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, N)}.$$

5. Let i_5 represent the mean waiting time of the orbital customer, which is defined by

$$i_5 = \frac{3}{4}.$$

6. The expected number of customers in the queue (i_6) of the multi-server queueing-inventory system in the steady-state is defined by

$$i_6 = \sum_{i_1=0} \dot{E} \sum_{i_6=1} \ddot{E} i_6 \Phi^{(i_1, 0, c, 0, 0, i_6)} + \sum_{i_1=0} \dot{E} \sum_{i_6=1} \ddot{E} i_6 \Phi^{(i_1, Q, c, 0, 0, i_6)} + \sum_{i_1=0} \dot{E} \sum_{i_2=0} \ddot{E} \sum_{i_3=0} \ddot{E} \sum_{i_6=c-i_3} \ddot{E} i_6 \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_6), i_6)}$$

$$\begin{aligned}
& + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_6 \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} \\
& + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_6 \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} \\
& + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_6 \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} \\
& + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_6 \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

7. let γ represent the expected rate at which the customer enters the waiting hall and is defined by

$$\begin{aligned}
\gamma = & \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, 0, c, 0, 0, i_6)} + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, Q, c, 0, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

8. Suppose that the queue is full. Any arriving primary customer can be lost with a probability of $1 - p$. The expected customer loss rate in the queue (ρ_9) of the multi-server queueing-inventory system in the steady state is defined as

$$\begin{aligned}
\rho_9 = & (1 - p) \lambda \Phi^{(i_1, 0, c, 0, 0, N)} + \\
& (1 - p) \lambda \Phi^{(i_1, Q, c, 0, 0, N)} + \\
& (1 - p) \lambda \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), N)} + \\
& (1 - p) \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, N)} + \\
& (1 - p) \lambda \Phi^{(i_1, i_2, i_3, c-i_3, 0, N)}.
\end{aligned}$$

9. In the steady state of the multi-server queueing-inventory system, the expected number of busy servers, denoted by ρ_{10} , and is defined by using the vector Φ as follows:

$$\begin{aligned}
\rho_{10} = & \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_4 \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_4 \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} (c - i_3) \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} +
\end{aligned}$$

$$\begin{aligned}
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_4 \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} (c - i_3) \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

10. While completing the service or vacation, if there are no items (excluding servicing items) and customers (excluding servicing customers) in the system, then the server takes a vacation. In the steady state of the multi-server queueing-inventory system, the expected number of servers on vacation, denoted by ρ_{11} , and is defined by using the vector Φ as follows:

$$\begin{aligned}
\rho_{11} = & \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_4 \Phi^{(i_1, 0, c, 0, 0, i_6)} + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_4 \Phi^{(i_1, Q, c, 0, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_3 \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_3 \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_3 \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_3 \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_3 \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)}.
\end{aligned}$$

11. At the end of the vacation or service, if the server is neither busy nor on vacation, the server stays in idle mode. Therefore, the expected number of idle servers ρ_{12} is defined by

$$\rho_{12} = c - (\rho_{10} + \rho_{11}).$$

12. The overall rate of retrial (ρ_{13}) of the stochastic multi-server queueing-inventory system in the steady-state is defined by

$$\rho_{13} = \theta \rho_3.$$

13. The retrial rate is successful when the number of customers in the queue is less than N . The successful rate of retrial is denoted by ρ_{14} and it is defined as:

$$\begin{aligned}
\rho_{14} = & \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, 0, c, 0, 0, i_6)} + \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, Q, c, 0, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, i_2, i_3, i_2, c-(i_3+i_2), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_4), i_4)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, 0, c, 0, 0, i_6)} + \\
& \frac{\rho_1 \rho_2 \rho_3}{\rho_1 \rho_2 \rho_3} \rho_1 \theta \Phi^{(i_1, Q, c, 0, 0, i_6)} +
\end{aligned}$$

$$\begin{aligned}
& \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, i_3, i_4, c-(i_3+i_6), i_6)} + \\
& i_1 = M+1 \quad i_2 = 0 \quad i_3 = 0 \quad i_6 = c-i_3 \\
& \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} + \\
& i_1 = M+1 \quad i_2 = 1 \quad i_3 = c-i_2 \quad i_6 = 0 \\
& \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& i_1 = M+1 \quad i_2 = 0 \quad i_3 = c-i_2 \quad i_6 = c-i_3+1 \\
& \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, i_3, i_6, c-(i_3+i_6), i_6)} + \\
& i_1 = M+1 \quad i_2 = c \quad i_3 = 0 \quad i_6 = 0 \\
& \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, i_3, c-i_3, 0, i_6)} + \\
& i_1 = M+1 \quad i_2 = c \quad i_3 = 0 \quad i_6 = c-i_3+1
\end{aligned}$$

14. The fraction of successful rate of retrial (ρ_{15}) is defined as the ratio of the overall rate of retrial to the successful retrial and is given by

$$\rho_{15} = \frac{14}{13}.$$

15. The probability of all servers being on vacation mode is defined as

$$\rho_{16} = \frac{c}{\lambda} \frac{c-1}{\lambda} \frac{c-2}{\lambda} \dots \frac{c-(\lambda-1)}{\lambda} \frac{N-1}{\lambda} \Phi^{(i_1, i_2, c, 0, 0, i_6)}.$$

5. Cost analysis and numerical illustration

The expected total cost (ETC) of the proposed model is given by

$$ETC = v_1 * \rho_{11} + v_2 * \rho_{12} + v_3 * \rho_{13} + v_4 * \rho_{14} + v_5 * \rho_{15}.$$

5.1. Parameter analysis

As per the result obtained in the stability condition and normalizing property, the following parameters and costs of the Markov process are assumed: $S = 32$, $s = 10$, $v_1 = 0.01$, $v_2 = 3$, $v_3 = 1$, $v_4 = 1.3$, $v_5 = 0.01$, $c = 3$, $Q = S - s$, $N = 4$, $M = 55$, $p = 0.7$, $\theta = 0.7$, $\lambda = 2.5$, $\mu = 5$, $\eta = 2.7$, $\beta = 1.5$, for the analysis of the numerical discussions. The impact of the parameters on the total cost, customer loss, busy servers, fraction of successful retrial rate, and waiting time are studied.

Example 1. In this example, the impact of the parameters λ , μ , θ , β , η , and c on the total cost can be observed in Table 1.

1. It has been noted that as the parameter value λ increases, the total cost also increases. This is due to the fact that the arrival rate increases, resulting in a greater ρ_{16} . Therefore, increasing the parameter value λ leads to an increase in the total cost.
2. As the θ increases, the total cost also increases. This is because more customers enter the waiting hall when the retrial rate increases. As a result, the product of cw and ρ_{16} is increased, leading to an increase in total cost.
3. If μ increases, then the total cost decreases. This is because an increase in μ leads to a decrease in the average service time per customer and a reduction in the size of the waiting room.
4. In Table 1 shows that as soon as vacation completion occurs, the customer waits in the queue also reduces quickly. Thus, it reduces the total cost.
5. While increasing the c in the system, the expected total cost of the system is also increased.
6. The expected total cost drops whenever the β increases because the replenishment time is decreasing.

Insights: Every business's primary goal is to reduce total costs. In such a way, this example investigates the total cost of changing various parameters and their characteristics. The parameter analysis provides insight to business people, allowing them to develop new strategies and ideas.

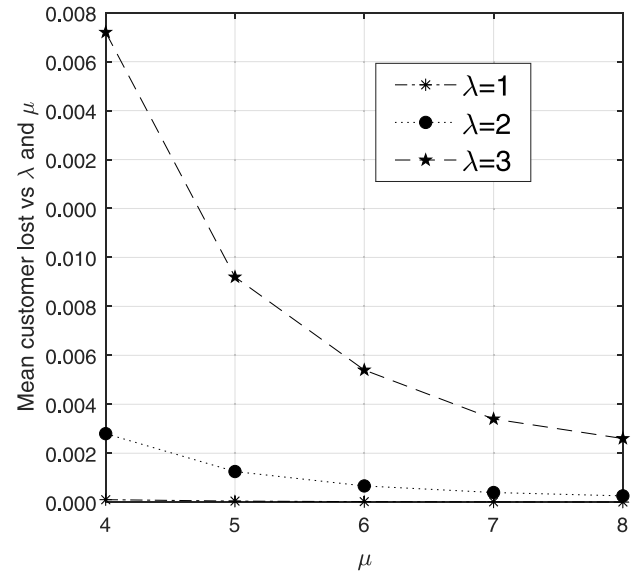


Fig. 2. Mean customer lost vs λ and μ .

Example 2. In this example, we study the impact of the expected customer loss rate by varying the parameters λ , μ , θ , β , η , and c in Figs. 2 and 4.

1. Whenever the λ , and θ increase, the mean number of customers lost also increases due to waiting hall overflow.
2. As the parameter μ grows, there is a corresponding decrease in the expected customer loss rate. As the service time per customer reduces, ρ_{16} will also drop concurrently.
3. When the rate η increases, the value of ρ_{15} decreases. When the server finishes his vacation, he immediately starts providing service. Consequently, customers in the waiting hall do not have to wait for an extended period, and the waiting area remains uncluttered. As a result, ρ_{15} is effectively managed.
4. The gradual decrease happens in the mean inventory level while increasing β . It ensures that the system has sufficient stocks in the inventory. Therefore, the customer in the waiting hall does not need to wait due to the inventory shortage, so the expected customer loss rate will be reduced.
5. As we stated for the parameter μ , increasing c reduces ρ_{16} quickly. So the overflow in the waiting room is then reduced, and the expected loss rate decreases automatically.

Insights: Customers are usually vital to any business. In business, the profit or growth is determined by the customer's arrival rate and purchase rate. When the arrival rate of a customer decreases, the sales process and profit are reduced. So a good administration always tries to reduce the customer loss rate of the business. In such a way, the above example gives the customer loss rate-related results to the readers (see Figs. 3 and 5).

Example 3. Figs. 6 and 7 explain the fraction of the successful retrial rate under the parameter variation.

1. As the rate λ increases, the fraction of the successful retrial rate decreases due to the overflow in the waiting hall.
2. The fraction of successful retrials increases due to the decrease of customers in the waiting hall while increasing the service rate μ . Similarly, the increase in the number of servers in the system causes an increase in the successful rate of retrials.

Table 1
Expected total cost under the influence of parameters.

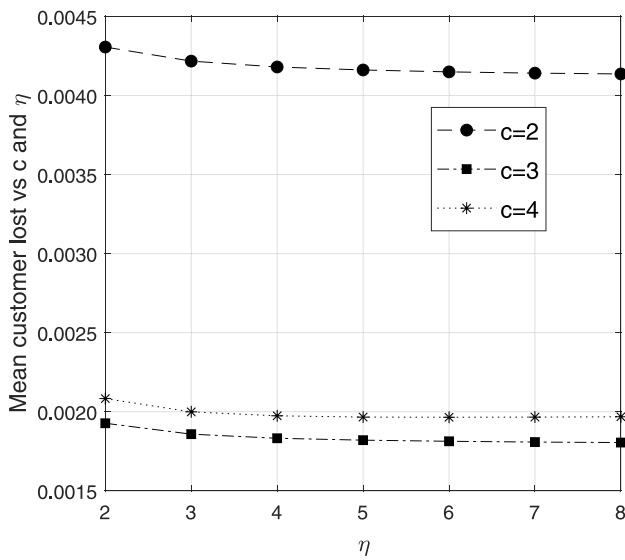
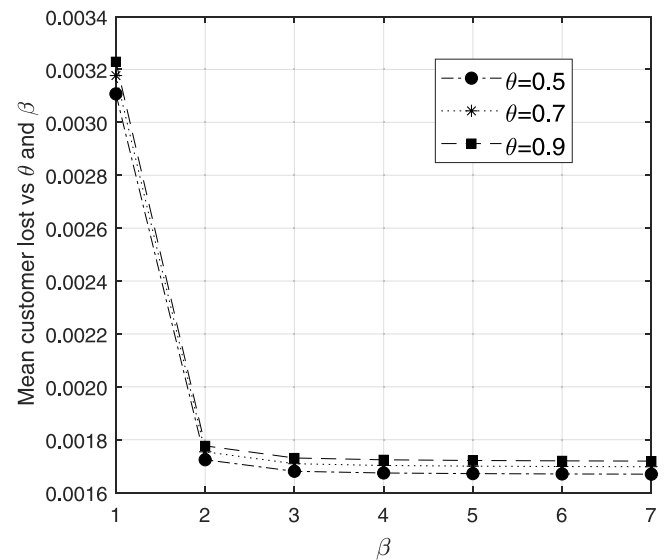
c	η	θ	β	λ	2.4	2.5			2.6					
					μ	4.9	5	5.1	4.9	5	5.1	4.9	5	5.1
2	2.6	0.6	1	3.606676	3.593941	3.581558	3.614909	3.601799	3.589070	3.625045	3.611544	3.598455		
			1.5	3.477352	3.462076	3.447430	3.492117	3.476244	3.461000	3.507800	3.491344	3.475522		
			2	3.451354	3.432587	3.414337	3.469717	3.450710	3.432218	3.488554	3.469285	3.450532		
		0.7	1	3.779718	3.767147	3.754890	3.785414	3.772513	3.759951	3.792931	3.779690	3.766815		
			1.5	3.666558	3.652014	3.638002	3.678276	3.663161	3.648577	3.690855	3.675189	3.660055		
			2	3.645263	3.627694	3.610569	3.660726	3.642942	3.625598	3.676603	3.658586	3.641005		
		0.8	1	3.927362	3.915031	3.902979	3.931186	3.918557	3.906230	3.936739	3.923808	3.911201		
			1.5	3.829858	3.816040	3.802668	3.839177	3.824818	3.810902	3.849308	3.834427	3.819990		
			2	3.813657	3.797316	3.781352	3.826740	3.810200	3.794031	3.840192	3.823437	3.807047		
	2.7	0.6	1	3.606572	3.593830	3.581441	3.614731	3.601613	3.588876	3.624792	3.611281	3.598182		
			1.5	3.476741	3.461416	3.446722	3.491491	3.475566	3.460272	3.507156	3.490646	3.474770		
			2	3.451015	3.432191	3.413879	3.469374	3.450311	3.431757	3.488208	3.468884	3.450069		
		0.7	1	3.779650	3.767070	3.754804	3.785278	3.772366	3.759794	3.792724	3.779471	3.766584		
			1.5	3.666003	3.651413	3.637357	3.677709	3.662547	3.647917	3.690273	3.674558	3.659375		
			2	3.644946	3.627322	3.610137	3.660408	3.642570	3.625167	3.676285	3.658214	3.640574		
		0.8	1	3.927320	3.914979	3.902916	3.931081	3.918441	3.906102	3.936568	3.923625	3.911004		
			1.5	3.829357	3.815497	3.802084	3.838667	3.824265	3.810306	3.848785	3.833859	3.819376		
			2	3.813366	3.796973	3.780952	3.826450	3.809860	3.793634	3.839903	3.823099	3.806654		
	2.8	0.6	1	3.606553	3.593807	3.581413	3.614648	3.601524	3.588782	3.624642	3.611125	3.598019		
			1.5	3.476312	3.460954	3.446227	3.491046	3.475086	3.459756	3.506692	3.490144	3.474230		
			2	3.450769	3.431903	3.413544	3.469126	3.450021	3.431420	3.487956	3.468590	3.449729		
		0.7	1	3.614648	3.601524	3.588782	3.785231	3.772313	3.759733	3.792616	3.779355	3.766459		
			1.5	3.491046	3.475086	3.459756	3.677319	3.662124	3.647462	3.689867	3.674117	3.658900		
			2	3.469126	3.450021	3.431420	3.660183	3.642306	3.624858	3.676059	3.657949	3.640266		
0.8		1	3.927355	3.915007	3.902936	3.931062	3.918414	3.906065	3.936493	3.923540	3.910908			
		1.5	3.829025	3.815139	3.801699	3.838327	3.823895	3.809907	3.848432	3.833475	3.818961			
		2	3.813164	3.796734	3.780671	3.826250	3.809624	3.793357	3.839705	3.822865	3.806379			
3	2.6	0.6	1	3.495840	3.484409	3.473384	3.500221	3.488523	3.477242	3.506351	3.494368	3.482813		
			1.5	3.357730	3.344500	3.331632	3.364837	3.351122	3.337809	3.373288	3.359116	3.345384		
			2	3.291876	3.276215	3.261216	3.306218	3.290006	3.274450	3.320846	3.304089	3.287985		
		0.7	1	3.666840	3.655528	3.644620	3.669181	3.657633	3.646501	3.673235	3.661439	3.650067		
			1.5	3.546744	3.533288	3.520174	3.551094	3.537171	3.523629	3.556768	3.542409	3.528467		
			2	3.496823	3.481235	3.466230	3.508170	3.492029	3.476464	3.519795	3.503108	3.486996		
		0.8	1	3.813479	3.802309	3.791538	3.814312	3.802929	3.791953	3.816807	3.805198	3.794007		
			1.5	3.710925	3.697343	3.684088	3.713207	3.699178	3.685513	3.716774	3.702327	3.688279		
			2	3.677404	3.661901	3.646906	3.686242	3.670190	3.654642	3.695366	3.678776	3.662688		
	2.7	0.6	1	3.494921	3.483473	3.472433	3.499163	3.487446	3.476149	3.505149	3.493148	3.481577		
			1.5	3.355157	3.341851	3.328910	3.362139	3.348339	3.334943	3.370463	3.356196	3.342373		
			2	3.288285	3.272284	3.256945	3.302647	3.286099	3.270207	3.317297	3.300207	3.283769		
		0.7	1	3.665907	3.654571	3.643642	3.668112	3.656539	3.645384	3.672027	3.660204	3.648809		
			1.5	3.544326	3.530787	3.517591	3.548558	3.534542	3.520909	3.554110	3.539648	3.525607		
			2	3.493458	3.477539	3.462202	3.504833	3.488364	3.472470	3.516486	3.499474	3.483034		
		0.8	1	3.812543	3.801343	3.790545	3.813244	3.801829	3.790825	3.815605	3.803963	3.792742		
			1.5	3.708662	3.694990	3.681647	3.710833	3.696705	3.682942	3.714284	3.699729	3.685575		
			2	3.674322	3.658501	3.643183	3.683191	3.666823	3.650954	3.692346	3.675442	3.659035		
	2.8	0.6	1	3.494242	3.482778	3.471725	3.498362	3.486629	3.475318	3.504223	3.492204	3.480618		
			1.5	3.353209	3.339844	3.326844	3.360072	3.346202	3.332740	3.368272	3.353927	3.340030		
			2	3.285439	3.269159	3.253538	3.299809	3.282984	3.266811	3.314469	3.297102	3.280384		
		0.7	1	3.665217	3.653858	3.642909	3.667302	3.655707	3.644531	3.671096	3.659249	3.647832		
			1.5	3.542517	3.528910	3.515650	3.546635	3.532544	3.518837	3.552070	3.537524	3.523401		
			2	3.490818	3.474627	3.459015	3.502209	3.485470	3.469302	3.513879	3.496597	3.479883		
0.8		1	3.811848	3.800621	3.789798	3.812435	3.800992	3.789963	3.814677	3.803006	3.791759			
		1.5	3.706987	3.693242	3.679828	3.709053	3.694843	3.681001	3.712393	3.697749	3.683507			
		2	3.671934	3.655851	3.640267	3.680824	3.664195	3.648060	3.689998	3.672833	3.656160			
0.6	1	3.532778	3.523127	3.513770	3.538067	3.528204	3.518643	3.544878	3.534795	3.525022				
	1.5	3.378662	3.368392	3.358487	3.386046	3.375645	3.365618	3.394580	3.384033	3.373869				
	2	3.280014	3.268435	3.257353	3.290207	3.278455	3.267227	3.301371	3.289434	3.278047				

2.6

(continued on next page)

Table 1 (continued).

c	η	θ	β	λ	2.4			2.5			2.6		
				μ	4.9	5	5.1	4.9	5	5.1	4.9	5	5.1
2.7	0.7	1	1	3.699196		3.690097	3.681278	3.702556	3.693276	3.684281	3.707354	3.697885	3.688708
		1.5		3.550242		3.540161	3.530452	3.555321	3.545134	3.535329	3.561550	3.551244	3.541328
		2		3.459053		3.447103	3.435680	3.466612	3.454506	3.442952	3.475169	3.462896	3.451202
		1		3.840692		3.832085	3.823744	3.842633	3.833870	3.825378	3.845934	3.837006	3.828356
		1.5		3.698239		3.688337	3.678814	3.701546	3.691554	3.681949	3.705981	3.695887	3.686188
		2		3.615313		3.603053	3.591345	3.620811	3.608403	3.596575	3.627308	3.614744	3.602785
	0.6	1		3.529149		3.519426	3.510002	3.534325	3.524387	3.514755	3.541034	3.530873	3.521026
		1.5		3.373016		3.362585	3.352527	3.380228	3.369663	3.359480	3.388619	3.377904	3.367579
		2		3.272382		3.260484	3.249095	3.282406	3.270329	3.258788	3.293442	3.281174	3.269468
		1		3.695644		3.686468	3.677574	3.698893	3.689532	3.680460	3.703593	3.694039	3.684782
		1.5		3.544947		3.534698	3.524829	3.549848	3.539489	3.529519	3.555925	3.545442	3.535357
		2		3.452030		3.439766	3.428038	3.459410	3.446981	3.435115	3.467824	3.455221	3.443207
	0.8	1		3.837260		3.828571	3.820151	3.839095	3.830245	3.821671	3.842301	3.833283	3.824548
		1.5		3.693320		3.683246	3.673559	3.696448	3.686280	3.676507	3.700728	3.690453	3.680582
		2		3.608889		3.596320	3.584313	3.614206	3.601482	3.589345	3.620555	3.607666	3.595391
		1		3.525914		3.516126	3.506639	3.530981	3.520975	3.511277	3.537593	3.527360	3.517444
		1.5		3.368033		3.357455	3.347256	3.375076	3.364359	3.354030	3.383320	3.372449	3.361976
		2		3.265737		3.253551	3.241883	3.275577	3.263203	3.251375	3.286464	3.273892	3.261893
2.8	0.7	1		3.692463		3.683215	3.674253	3.695606	3.686171	3.677028	3.700211	3.690580	3.681249
		1.5		3.540272		3.529868	3.519851	3.544999	3.534480	3.524358	3.550923	3.540276	3.530035
		2		3.445941		3.433392	3.421386	3.453130	3.440406	3.428254	3.461384	3.448478	3.436172
	0.8	1		3.834176		3.825410	3.816917	3.835908	3.826979	3.818329	3.839023	3.829923	3.821108
		1.5		3.688979		3.678747	3.668907	3.691933	3.681602	3.671671	3.696059	3.685616	3.675583
		2		3.603344		3.590496	3.578217	3.608472	3.595458	3.583040	3.614658	3.601472	3.588908

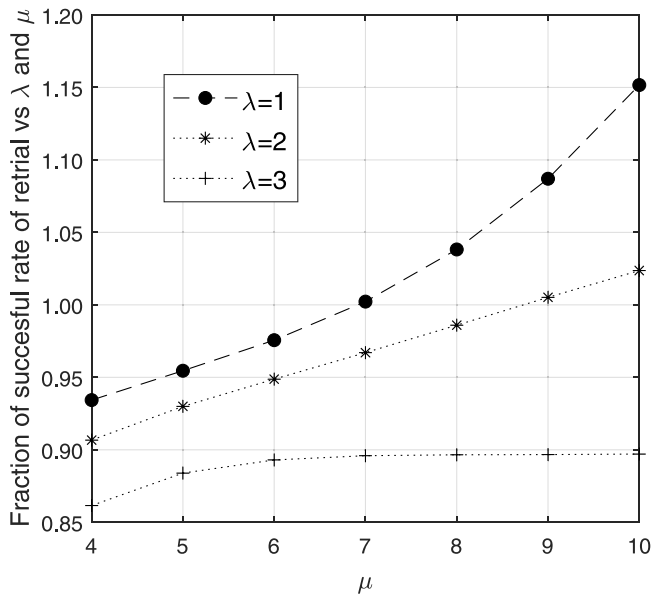
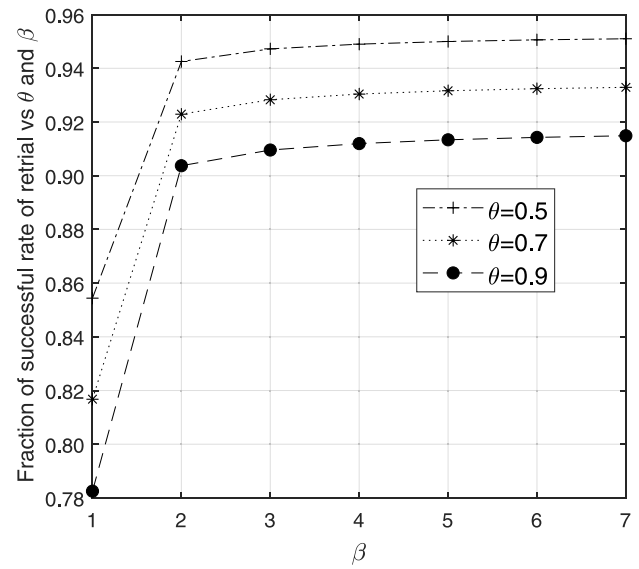
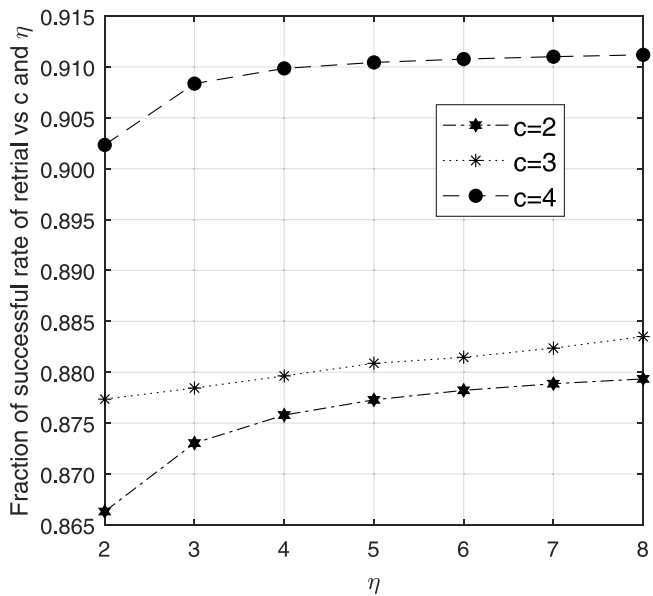
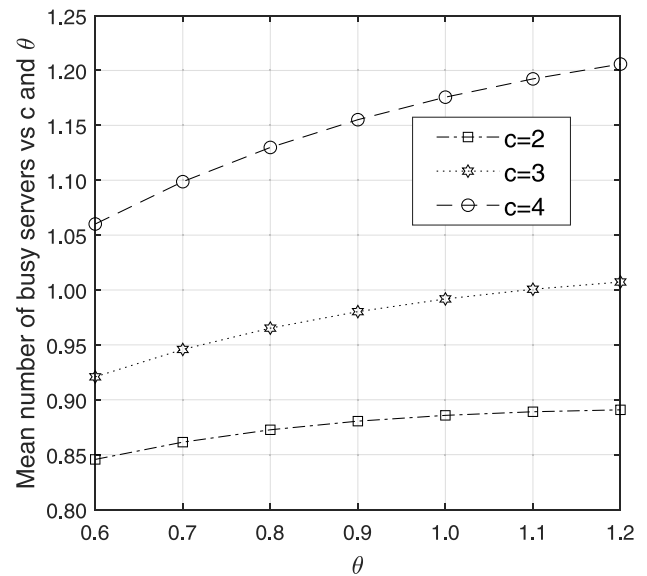
Fig. 3. Mean customer lost vs c and η .Fig. 4. Mean customer lost vs θ and β .

- As we predicted, the rate θ reduces the fraction of successful retrials when it increases because more customers from the orbit enter the waiting hall.
- When β increases, the inventory storage system maintains a positive stock level so that an arriving customer can purchase the item and leave the system without delay. Hence, it increases the fraction of the successful retrial rate.
- Suppose the server returns from vacation sooner than expected, then many customers get their service without a server delay. On the other hand, if the server is on vacation, a customer must wait until his turn occurs. So, if the servers are already available in the system, his turn occurs quickly. Therefore, the vacation completion rate increases the fraction of the success rate of a retrial if it increases.

Insights: When we approach a business economically, caring for the orbital customer's arrival will influence the growth of the business. So, this example explores the characteristics of the fraction of successful retrial rates of orbit customers with parameter variation. In such a way, the parameters c , β , and η are important in overcoming the loss of retrial customers.

Example 4. In this example, we explained the impact of the parameters λ , θ , μ , c , β and η on the mean number of servers on vacation and the mean number of busy servers, as shown in Figs. 8–13.

- As depicted in Fig. 10, an increase in the rate λ is associated with a decrease in the average number of servers on vacation. The prohibition of vacation continuation is due to an adequate number of clients in the queue. As the parameter μ increases, there is a corresponding increase in the value of η . Because

Fig. 5. Fraction of successful rate of retrial vs λ and μ .Fig. 7. Fraction of successful rate of retrial vs θ and β .Fig. 6. Fraction of successful rate of retrial vs c and η .Fig. 8. Mean number of busy servers vs c and θ .

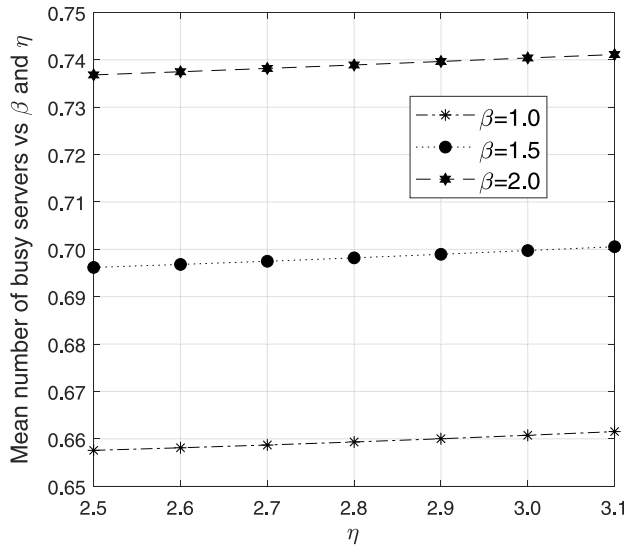
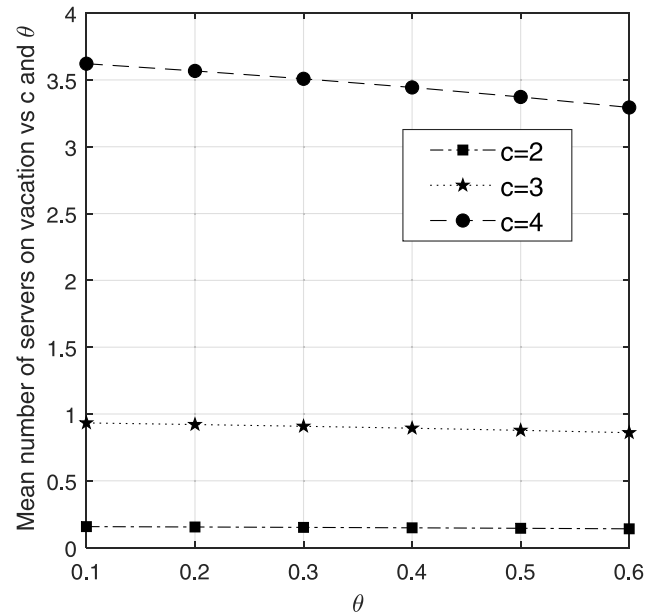
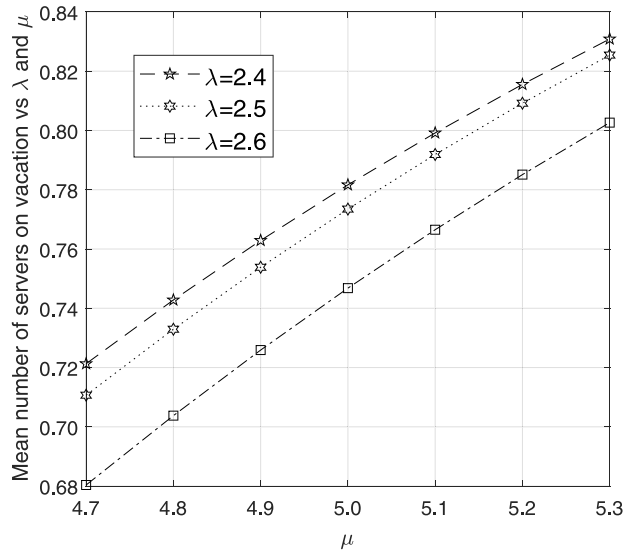
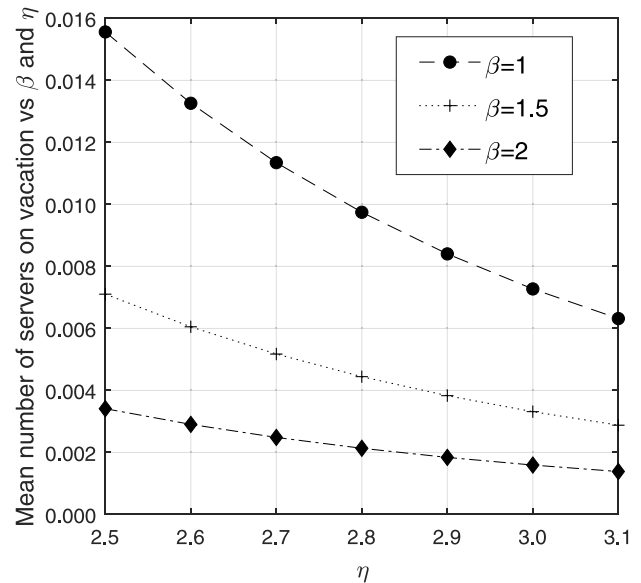
the service duration per customer decreases in response to the increment of μ .

- In Fig. 11, if the change in θ is small, then the change in π_{11} is also small with an indirect variation. For the fixed waiting hall size, when $c = 2$ and $c = 3$, the change in π_{11} is small, but when $c = 4$, it is to be noted that the change in π_{11} is high.
- As depicted in Fig. 12, mean number of servers are in vacation reduces when β increases. If the replenishment has been done quickly, the server cannot take a vacation. By increasing η , the amount of time a server spends on vacation is reduced. Therefore, the server must frequently return to the system, which confirms the decrease in the number of servers on vacation.
- Fig. 13 shows that the rate μ decreases the π_{10} if it increases. It helps to control the crowd in the system. As the number of

customers in the queue shortly decreases, the number of busy servers will also be reduced. The π_{10} will be increased as λ increases. Because of the increase in arrivals, all ideal servers will be activated instantly.

- If the parameters c and θ increase, then the mean number of busy servers increases simultaneously since the arrival of a retrial customer enters the queue and activates each server. Once the number of servers is increased, he will immediately join the system to provide customer service. This is shown in Fig. 8.
- In Fig. 9, β and η are directly proportional to π_{10} . If the change in η is small, then the π_{10} is also small.

Insights: Analyzing the mean number of servers on vacation and the mean number of busy servers can provide valuable insights for business professionals. By examining the impact of various parameters on π_{10} and π_{11} , readers can develop new strategies and ideas to reduce

Fig. 9. Mean number of busy servers vs β and η .Fig. 11. Mean number of servers on vacation vs c and θ .Fig. 10. Mean number of servers on vacation vs λ and μ .Fig. 12. Mean number of servers on vacation vs β and η .

the average number of servers on vacation and increase the average number of busy servers in their businesses.

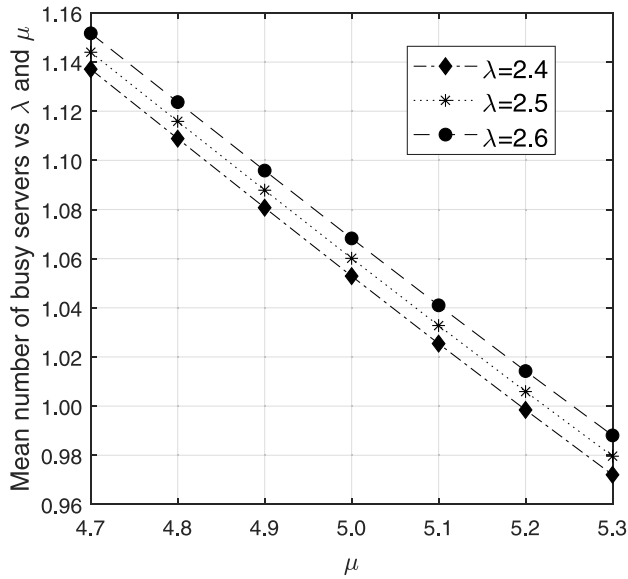
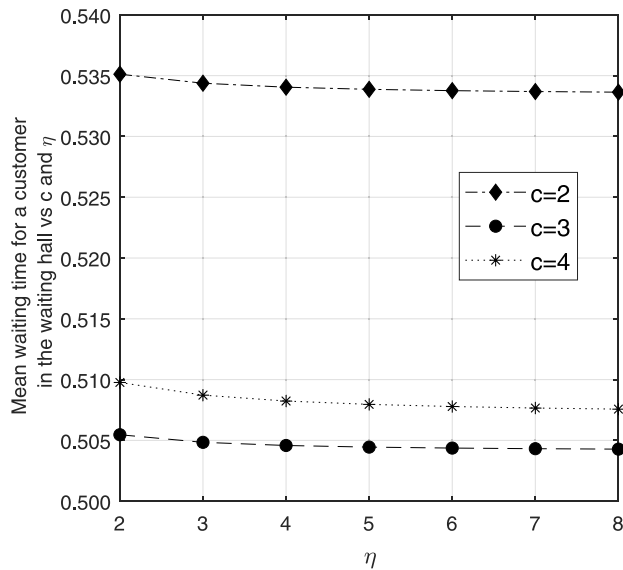
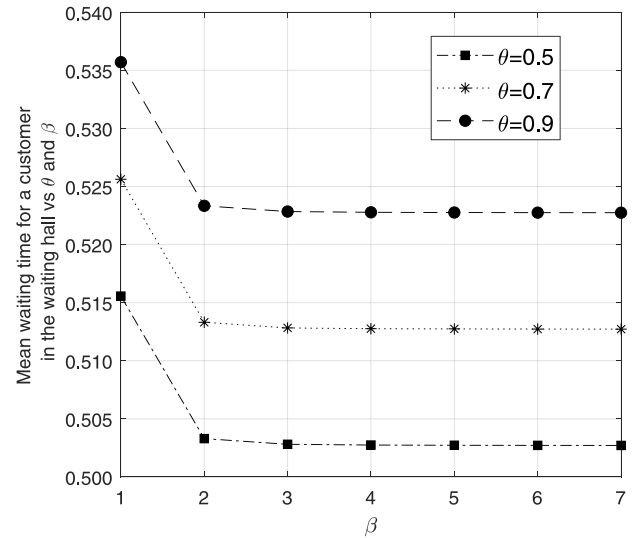
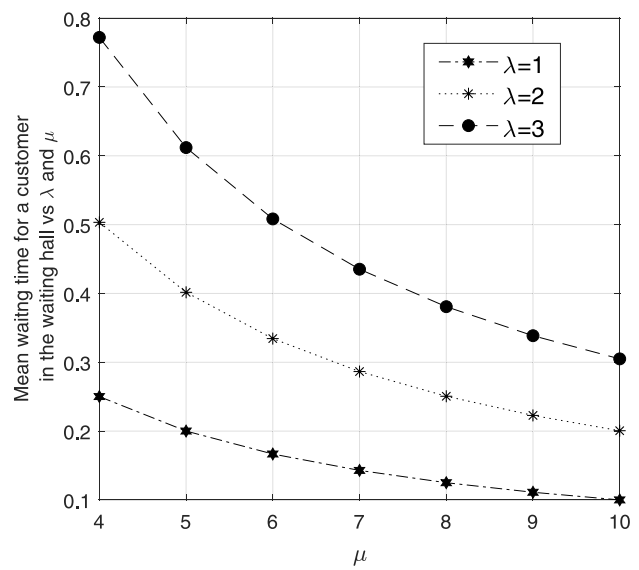
Example 5. This example examines the orbit and waiting hall customer's waiting times in Figs. 14–19 respectively, using the parameters λ , μ , θ , β , c , and η .

1. The parameters μ and c have an equivalent impact on reducing the client waiting time, whether they are in the queue or orbit. In other words, decreasing the average time it takes to complete a service or increasing the number of servers will decrease the number of customers waiting in the queue, which will also lead to a reduction in the number of customers in orbit. This ultimately leads to a reduction in the waiting time for both sets of customers.
2. In a similar way, the replenishment rate helps the system avoid a shortage of inventory if it increases. Since the stock is positive, the customers in the waiting hall do not need to wait a long time

in a system due to the lack of items. So, both customers' waiting times will be reduced.

3. The vacation completion rate also contributes to reducing the customers' waiting time in a queue or orbit. This rate ensures that the servers are primarily available when it increases.
4. If λ increases, the number of customers in the waiting hall increases and soon overflows. Thus, customers' waiting times in the queue and the orbit increase whenever it increases.
5. Increasing θ only affects the waiting time of customers in orbit, not those in the waiting hall. This is because the waiting hall quickly fills up, and the increments in θ result in a decrease in the orbit's level, which leads to a reduction in their waiting time.

Insights: Analyzing waiting time in customer-centric service systems to run a successful business is crucial. Ensuring customer satisfaction

Fig. 13. Mean number of busy servers vs λ and μ .Fig. 14. Mean waiting time for a customer in the waiting hall vs c and η .Fig. 15. Mean waiting time for a customer in the waiting hall vs β and θ .Fig. 16. Mean waiting time for a customer in the waiting hall vs λ and μ .

through the service system is paramount in this environment, as reducing customers' waiting time leads to higher satisfaction levels and a better reputation for the company. This, in turn, leads to more customers returning to make additional purchases within the same system. Therefore, this example is a persuasive argument for readers and business people to prioritize reducing customer waiting times.

6. Conclusion

This article investigated a multi-server queueing-inventory system with asynchronous multiple-server vacations. In addition, this model explored the retrial facility and (s, Q) reordering policy for the replenishment process. As per the assumptions, a six-dimensional stochastic process is framed and transformed into an infinitesimal matrix by the transition of states. The stability condition and joint probability vector are obtained through the Neuts and Rao's Truncation method based on the structure of the infinitesimal generator matrix. Moreover,

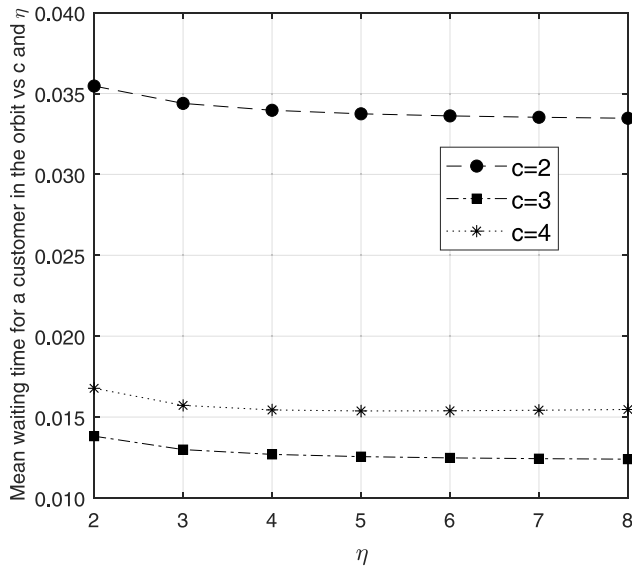
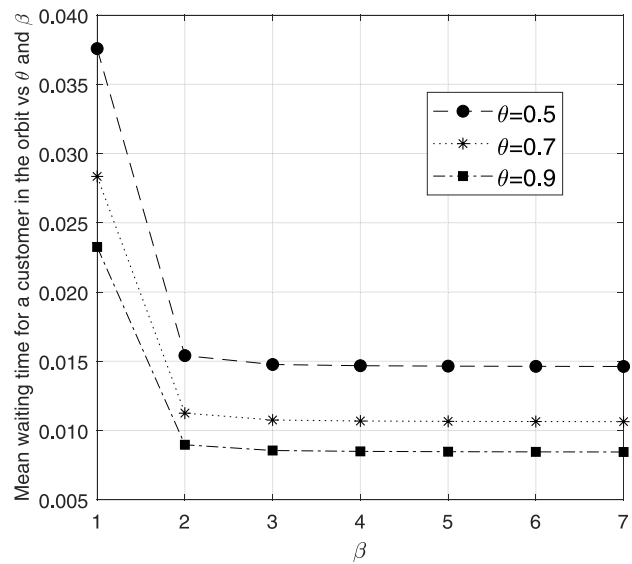
the expected total cost function is established by considering specific system performance measures.

From the analysis of this paper, we observe the following results:

6.1. Observations

The numerical investigations obtain these observations:

1. From the parameter analysis, β , η , and μ are directly proportional to an expected total cost, whereas λ , θ and c are inversely proportional to the expected total cost.
2. The parameters β , η , and μ contribute to giving a rest period to the servers and react with an indirect variation.
3. The expected loss rate of a customer is controlled by the increase of the parameters β , η , c , and μ .
4. The fraction of the successful retrial rate of an orbit customer is slowed down due to the overflow of the waiting hall. To control

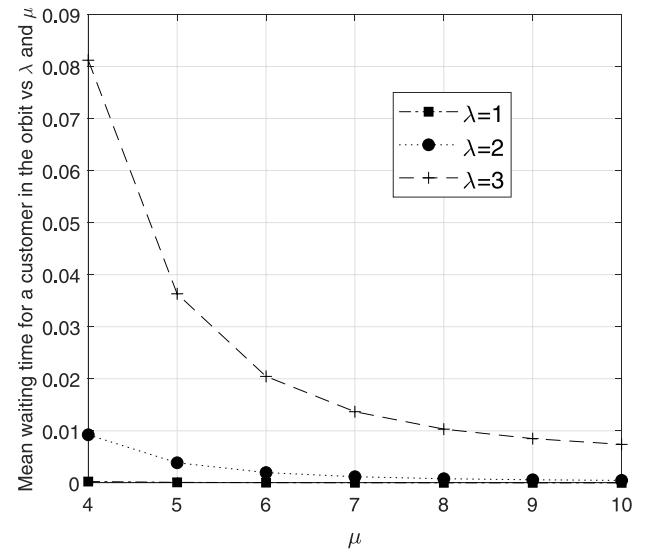
Fig. 17. Mean waiting time for a customer in the orbit vs c and η .Fig. 18. Mean waiting time for a customer in the orbit vs θ and β .

such an overflow, one can increase the values of parameters β , η , c , and μ as required.

5. The waiting time of queue and orbit customers has an indirect variation relation with the parameters β , η , c , and μ . As the situation requires, one can vary the parameter value to get the desired output.

6.2. Limitations

When considering $c = 1$, this model converts into a single server queueing-inventory system with multiple server vacations. If the retrial customer is independent of the customer in orbit, then the corresponding classical retrial policy is converted to a constant retrial policy. Assuming $p = 1$, the proposed system sends the overflowing customers to an orbit compulsorily, whereas if we assume $p = 0$, all the customers are considered lost.

Fig. 19. Mean waiting time for a customer in the orbit vs λ and μ .

6.3. Future direction

The proposed model can be extended into a multi-server retrial queueing-inventory system in which customers arrive according to a Markovian arrival process (MAP) and the service times to be of phase type.

Declaration of competing interest

The authors of the paper declare that they have no conflict of interest.

Data availability

No data was used for the research described in the article.

Appendix A

$$J_{0^*} = \begin{cases} \eta, & i_3 = c-1, & i_4 = 1, & i_5 = 0, & i_6 \in \Xi_1^N \\ i_3' = c, & i_4' = 0, & i_5' = i_5, & i_6' = i_6 - 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$J_{Q^*} = \begin{cases} \beta, & i_3 = c, & i_4 = 0, & i_5 = 0, & i_6 \in \Xi_0^N \\ i_3' = c, & i_4' = i_4, & i_5' = i_5, & i_6' = i_6, \\ 0, & \text{otherwise.} \end{cases}$$

$$J_0 = \begin{cases} \eta, & i_3 = c, & i_4 = 0, & i_5 = 0, & i_6 \in \Xi_1^N \\ i_3' = c-1, & i_4' = i_4, & i_5' = 1, & i_6' = i_6, \\ 0, & \text{otherwise.} \end{cases}$$

$$J_Q = \begin{cases} \eta, & i_3 = c, & i_4 = 0, & i_5 = 0, & i_6 \in \Xi_0^N \\ i_3' = c-1, & i_4' = 1-i_5', & i_5' = \delta_0' 0, & i_6' = i_6, \\ 0, & \text{otherwise.} \end{cases}$$

For $t_2 \in \Xi_1^{c-1}$

$$F_{t_2} = \begin{cases} h & i_4\mu, & i_3 \in \Xi_0^{c-(t_2+1)}, & i_4 = i_2, \\ n & i'_3 = i_3 + 1, & i'_4 = i_4 - 1, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4 + i_5, \\ n & i'_5 = i_5, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 \in \Xi_1^{c-(t_2+1)}, & i_4 = i_2, \\ n & i'_3 = i_3, & i'_4 = i_4 - 1, \\ n & i_5 = c - (i_3 + i_4), & i_6 \in \Xi_{i_4+i_5+1}^N, \\ n & i'_5 = i_5 + 1, & i'_6 = i_6 - 1, \\ n & \bar{\delta}_{i_2 1} i_4\mu, & i_3 \in \Xi_{c-i_2}^{c-2}, & i_4 \in \Xi_1^{c-i_3}, \\ n & i'_3 = i_3 + 1, & i'_4 = i_4 - 1, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & i'_5 = i_5, & i'_6 = i_6 - 1, \\ n & \bar{\delta}_{i_2 1} i_4\mu, & i_3 \in \Xi_{c-i_2}^{c-2}, & i_4 = i_2, \\ n & i'_3 = i_3, & i'_4 = i_4 - 1, \\ n & i_5 = 0, & i_6 \in \Xi_{c-i_3}^N, \\ n & i'_5 = i_5 + 1, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 = c - 1, & i_4 = 1, \\ n & i'_3 = i_3, & i'_4 = \delta_{i_6 2}, \\ n & i_5 = 0, & i_6 \in \Xi_1^N, \\ n & i'_5 = \delta_{i_6 1}, & i'_6 = i_6 - 1, \\ n & 0, & \text{otherwise.} \end{cases}$$

For $t_2 \in \Xi_0^{c-1}$

$$L_{t_2} = \begin{cases} h & \beta, & i_3 \in \Xi_0^{c-(t_2+1)}, & i_4 = i_2, \\ n & i'_3 = i_3, & i'_4 = c - i'_3, \\ n & i_5 = c - (i_3 + i_4), & i_6 \in \Xi_{i_4+i_5}^N, \\ n & i'_5 = 0, & i'_6 = i_6, \\ n & \bar{\delta}_{i_2 0} \beta, & i_3 \in \Xi_{c-i_2}^{c-1}, & i_4 \in \Xi_0^{c-i_3}, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & i'_5 = i_5, & i'_6 = i_6, \\ n & \bar{\delta}_{i_2 0} \beta, & i_3 \in \Xi_{c-i_2}^{c-1}, & i_4 = i_2, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = 0, & i_6 \in \Xi_{c-i_3+1}^N, \\ n & i'_5 = i_5, & i'_6 = i_6, \\ n & 0, & \text{otherwise.} \end{cases}$$

$$L_c = \begin{cases} h & \beta, & i_3 \in \Xi_0^{c-1}, & i_4 \in \Xi_0^{c-i_3}, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & i'_5 = i_5, & i'_6 = i_6, \\ n & \beta, & i_3 \in \Xi_0^{c-1}, & i_4 = c - i_3, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = 0, & i_6 \in \Xi_{i_4+1}^N, \\ n & i'_5 = i_5, & i'_6 = i_6, \\ n & 0, & \text{otherwise.} \end{cases}$$

Appendix B

$$F_c = \begin{cases} h & i_4\mu, & i_3 = 0, & i_4 \in \Xi_1^c, \\ n & i'_3 = 1, & i'_4 = i_4 - 1, \\ n & i_5 = c - i_4, & i_6 = i_4, \\ n & i'_5 = i_5 - 1, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 = 0, & i_4 = c, \\ n & i'_3 = i_3, & i'_4 = i_4 - 1, \\ n & i_5 = 0, & i_6 \in \Xi_{c+1}^N, \\ n & i'_5 = 1, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 \in \Xi_1^{c-1}, & i_4 \in \Xi_1^{c-i_3}, \\ n & i'_3 = i_3, & i'_4 = i_4 - 1, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & i'_5 = i_5 + 1, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 \in \Xi_1^{c-1}, & i_4 = c - i_3, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = 0, & i_6 \in \Xi_{i_4+1}^N, \\ n & i'_5 = i_5, & i'_6 = i_6 - 1, \\ n & 0, & \text{otherwise.} \end{cases}$$

For $t_2 \in \Xi_{c+1}^S$

$$F_{c+1} = \begin{cases} h & i_4\mu, & i_3 \in \Xi_0^{c-1}, & i_4 \in \Xi_1^{c-i_3}, \\ n & i'_3 = i_3, & i'_4 = i_4 - 1, \\ n & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & i'_5 = i_5 + 1, & i'_6 = i_6 - 1, \\ n & i_4\mu, & i_3 \in \Xi_0^{c-1}, & i_4 = c - i_3, \\ n & i'_3 = i_3, & i'_4 = i_4, \\ n & i_5 = 0, & i_6 \in \Xi_{i_4+1}^N, \\ n & i'_5 = i_5, & i'_6 = i_6 - 1, \\ n & 0, & \text{otherwise.} \end{cases}$$

$$G_{t_2} = \begin{cases} h & -(i_1\theta + \bar{\delta}_{i_6 N} \lambda + \delta_{i_2 0^*} \beta \\ n & + \delta_{i_2 0^*} \bar{\delta}_{i_6 N} \eta \\ n & + \delta_{i_2 Q^*} \eta + \delta_{i_6 N} p\lambda), & i_3 = c, & i_4 = 0, \\ n & & i'_3 = c, & i'_4 = i_4, \\ n & & i_5 = 0, & i_6 \in \Xi_0^N, \\ n & & i'_5 = i_5, & i'_6 = i_6, \\ n & \lambda, & i_3 = c, & i_4 = 0, \\ n & & i'_3 = c, & i'_4 = i_4, \\ n & & i_5 = 0, & i_6 \in \Xi_0^{N-1}, \\ n & & i'_5 = i_5, & i'_6 = i_6 + 1, \\ n & 0, & \text{otherwise.} \end{cases}$$

For $t_2 \in \Xi_0^{c-1}$

$$G_{t_2} = \begin{cases} h & \lambda, & i_3 \in \Xi_0^{c-(t_2+1)}, & i_4 = i_2, \\ n & & i'_3 = i_3, & i'_4 = i_4, \\ n & & i_5 = c - (i_3 + i_4), & i_6 \in \Xi_{i_4+i_5}^{N-1}, \\ n & & i'_5 = i_5, & i'_6 = i_6 + 1, \\ n & \bar{\delta}_{i_2 0} \lambda, & i_3 \in \Xi_{c-i_2}^{c-1}, & i_4 \in \Xi_0^{c-i_3}, \\ n & & i'_3 = i_3, & i'_4 = i_4, \\ n & & i_5 = c - (i_3 + i_4), & i_6 = i_4, \\ n & & i'_5 = i_5, & i'_6 = i_6 + 1, \\ n & \bar{\delta}_{i_2 0} \lambda, & i_3 \in \Xi_{c-i_2}^{c-1}, & i_4 = i_2, \\ n & & i'_3 = i_3, & i'_4 = i_4, \\ n & & i_5 = 0, & i_6 \in \Xi_{c-i_3+1}^{N-1}, \\ n & & i'_5 = i_5, & i'_6 = i_6 + 1, \\ n & -(i_4\mu \\ n & + \bar{\delta}_{i_6 N} (\lambda + i_1\theta) \\ n & + \bar{\delta}_{i_3 0} \bar{\delta}_{i_6 i_5} \eta \\ n & + \delta_{i_6 N} p\lambda), & i_3 \in \Xi_0^{c-(t_2+1)}, & i_4 = i_2, \\ n & & i'_3 = i_3, & i'_4 = i_4, \\ n & & i_5 = c - (i_3 + i_4), & i_6 \in \Xi_{i_4+i_5}^N, \\ n & & i'_5 = i_5, & i'_6 = i_6, \end{cases}$$

$$G_{i_2} = \begin{cases} \begin{aligned} & -\bar{\delta}_{i_2 0}(i_4 \mu + \\ & (\lambda + i_1 \theta) + \\ & \bar{\delta}_{i_2(c-1)} \bar{\delta}_{i_3(c-1)} \eta \\ & + \delta_{i_6 N} p \lambda), \end{aligned} & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_0^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6, \end{aligned} \\ \\ \begin{aligned} & -\bar{\delta}_{i_2 0}(i_4 \mu + \\ & \bar{\delta}_{i_6 N}(\lambda + i_1 \theta) \\ & + \eta + \delta_{i_6 N} p \lambda), \end{aligned} & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = 0, \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = i_2, \\ i'_4 & = i_4, \\ i_3 & \in \Xi_{i_2+1}^N, \\ i'_6 & = i_6, \end{aligned} \\ \\ 0, & \text{otherwise.} \end{cases}$$

For $i_2 \in \Xi_c^S$

$$G_c = \begin{cases} \begin{aligned} & -(i_4 \mu + \delta_{i_3 0} \eta \\ & + \lambda + i_1 \theta), \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_0^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6, \end{aligned} \\ \\ \begin{aligned} & -(i_4 \mu + \eta \\ & + \delta_{i_6 N}(\lambda + i_1 \theta) \\ & + \delta_{i_6 N} p \lambda), \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = 0, \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = c - i_3, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{c-i_3+1}^N, \\ i'_6 & = i_6, \end{aligned} \\ \\ \lambda, & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_c^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ \lambda, & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = 0, \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = c - i_3, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{c-i_3+1}^{N-1}, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ 0, & \text{otherwise.} \end{cases}$$

For $i_2 \in 0^*, Q^*$

$$\hat{G}_{i_2} = \begin{cases} \begin{aligned} & -(M\theta + \bar{\delta}_{i_6 N} \lambda + \\ & \bar{\delta}_{i_2 0^*} \beta + \delta_{i_2 0^*} \bar{\delta}_{i_6 N} \eta + \\ & \bar{\delta}_{i_2 Q^*} \eta) \end{aligned} & \begin{aligned} i_3 & = c, \\ i'_3 & = c \\ i_5 & = 0, \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = 0, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_0^N \\ i'_6 & = i_6 \end{aligned} \\ \\ \lambda + M\theta, & \begin{aligned} i_3 & = c, \\ i'_3 & = c, \\ i_5 & = 0 \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = 0, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_0^{N-1} \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ 0, & \text{otherwise.} \end{cases}$$

For $i_2 \in \Xi_0^{c-1}$

$$\hat{G}_{i_2} = \begin{cases} \begin{aligned} & \lambda + M\theta, \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-(i_2+1)}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = i_2, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{i_4+i_5}^{N-1}, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ \bar{\delta}_{i_2 0}(\lambda + M\theta), & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_0^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ \bar{\delta}_{i_2 0}(\lambda + M\theta) & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = 0, \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = i_2, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{c-i_3+1}^{N-1}, \\ i'_6 & = i_6 + 1 \end{aligned} \\ \\ \begin{aligned} & -(i_4 \mu \\ & + \bar{\delta}_{i_6 N}(\lambda + M\theta) \\ & + \bar{\delta}_{i_3 0} \bar{\delta}_{i_6 i_5} \eta) \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-(i_2+1)}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = i_2, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{i_4+i_5}^N, \\ i'_6 & = i_6 \end{aligned} \\ \\ \begin{aligned} & -\bar{\delta}_{i_2 0}(i_4 \mu + \\ & (\lambda + M\theta) + \\ & \bar{\delta}_{i_2(c-1)} \bar{\delta}_{i_3(c-1)} \eta) \end{aligned} & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = c - (i_3 + i_4) \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_0^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6, \end{aligned} \\ \\ \begin{aligned} & -\bar{\delta}_{i_2 0}(i_4 \mu + \\ & \bar{\delta}_{i_6 N}(\lambda + M\theta) \\ & + \eta), \end{aligned} & \begin{aligned} i_3 & \in \Xi_{c-i_2}^{c-1}, \\ i'_3 & = i_3, \\ i_5 & = 0, \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = i_2, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{i_2+1}^N, \\ i'_6 & = i_6 \end{aligned} \\ \\ 0, & \text{otherwise.} \end{cases}$$

For $i_2 \in \Xi_c^S$

$$\hat{G}_c = \begin{cases} \begin{aligned} & -(i_4 \mu + \delta_{i_3 0} \eta \\ & + \lambda + M\theta), \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3 \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & \in \Xi_0^{c-i_3}, \\ i'_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6, \end{aligned} \\ \\ \begin{aligned} & -(i_4 \mu + \eta \\ & + \bar{\delta}_{i_6 N}(\lambda + M\theta)) \end{aligned} & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3 \\ i_5 & = 0 \\ i'_5 & = i_5 \end{aligned} & \begin{aligned} i_4 & = c - i_3, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{c-i_3+1}^N \\ i'_6 & = i_6 \end{aligned} \\ \\ \lambda + M\theta & \begin{aligned} i_3 & = i_3 \\ i_5 & = c - (i_3 + i_4), \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = i_4, \\ i_6 & = i_4, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ \lambda + M\theta & \begin{aligned} i_3 & \in \Xi_0^{c-1}, \\ i'_3 & = i_3 \\ i_5 & = 0, \\ i'_5 & = i_5, \end{aligned} & \begin{aligned} i_4 & = c - i_3, \\ i'_4 & = i_4, \\ i_6 & \in \Xi_{c-i_3+1}^{N-1}, \\ i'_6 & = i_6 + 1, \end{aligned} \\ \\ 0, & \text{otherwise,} \end{cases}$$

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